OPEN PROM

Model Version 0.1

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created with goxygen on Sep 12 2023

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OPEN PROM

The OPEN PROM is a model to test and present the basic functionality of the gms and goxygen packages (@gms, @goxygen).

It has the structure of a plain GAMS model and can be interpreted by goxygen, but has no content and cannot be solved with GAMS.

*** preliminaries

*** number of columns that are listed for each variable in the column listing

Let's add some standard code

```
option limcol = 16;
```

*** number of rows that are listed for each equation in the equation listing

```
option limrow = 16;
```

*** "dollar" ($) commands section: define GAMS flags & code control & compilation-time options

*** onEolCom: turn on end-of-line comments (starting with !!, i.e. the GAMS default)

```
$onEolCom
```

*** onEnd: turn on alternative flow control syntax (more readable loop, for, if etc.)

```
$onEnd
```

*** onEmpty allows declarations of empty parameters

```
$onEmpty
```

*** TODO: check if the contents of this block are actually used later

*** GAMS “flags” definitions

*** Maximum number of solver attempts

```
$evalGlobal SolverTryMax 1
```

```
$setGlobal fCountries 'RAS,RWO,REU,REP,RAF,RLA,ARG,OCE,AUT,BEL,BRA,BGR,
    CAN,CHA,HRV,CYP,CZE,DNK,EGY,EST,FIN,FRA,DEU,GRC,HUN,NSI,IND,IDN,IRN,
    IRL,ISR,ITA,JPN,KOR,LVA,LTU,LUX,MLT,MEX,MAR,NLD,NGA,POL,PRT,ROU,RUS,
    SAU,SVK,SVN,ZAF,ESP,SWE,TUN,TUR,GBR,USA'
```

```
$setGlobal fSingleCountryRun 'yes'
```

```
$setGlobal fCountryList %countries%
```

```
$setGlobal fScenario 'DECARB_400'
$setGlobal fBaseline 'BASE'
```

```
$evalGlobal fPeriodOfYears 1
```

```
$setGlobal fIncludeNonCO2 no
```

```
$setGlobal fReadCommonDB 'yes'
$setGlobal fReadCountryDB 'yes'
$setGlobal fReadCountryCalib 'yes'
```

```
$evalGlobal fStartHorizon 2010
$evalGlobal fEndHorizon 2100
$evalGlobal fEndY 2020
$evalGlobal fStartY 2018
$evalGlobal fBaseY %fStartY% - %fPeriodOfYears%
```

*** end of dollar commands section, no further flag definitions allowed

*** load input data files

```
$include sets.gms
$include declarations.gms
$include input.gms
$include equations.gms
$include preloop.gms
$include solve.gms
```

Authors

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How to cite

Anastasis G, Georgios P (2019). “OPEN PROM - Version 0.1.”

Bibtex format

```bibtex
@Misc{,
  title = {OPEN PROM - Version 0.1},
  author = {Giannousakis Anastasis and Plessias Georgios},
  date = {2019-05-13},
  year = {2019},
}
```

Citation File Format

```cff
cff-version: 1.0.3
message: If you use this model, please cite it as below.
authors:
- family-names: Anastasis
  given-names: Giannousakis
  affiliation: E3Modelling
  email: bat@m.an
- family-names: Georgios
  given-names: Plessias
  affiliation: E3Modelling
```

```yaml
  email: bat@m.an
title: OPEN PROM
version: '0.1'
date-released: '2019-05-13'
repository-code: https://github.com/e3modelling/OPEN-PROM
keywords: testing
license: CC0
```

Equations (equations.gms)

Description

Declaration of variables and equation.

- Power Generation
- Compute current renewable potential

$$VCurrRenPot(runCy, PGRENEF, YTIME) = \frac{(VMaxmAllowRenPotent(runCy, PGRENEF, YTIME) + iMinRenPotential(runCy, PGRENEF, YTI}{2}$$

- Compute CHP electric capacity

$$VElecCapChpPla(runCy, CHP, YTIME) = \frac{1}{sTWhToMtoe} \cdot \sum_{INDDOM} VConsFuel(runCy, INDDOM, CHP, YTIME) \cdot \frac{VElecIndPrices(runCy,YTIME)}{\frac{\sum_{PGALL\$CHPtoEON(CHP,PGALL)} iAvailRate(PGALL,YTIME)}{\frac{iUtilRateChpPlants(runCy,CHP,YTIME)}{sGwToTwhPerYear}}}$$

- Compute Lambda parameter

$$\frac{(1 - exp\,(-VLoadCurveConstr(runCy, YTIME) \cdot sGwToTwhPerYear))}{VLoadCurveConstr(runCy, YTIME)} = \frac{(VElecDem(runCy, YTIME) - sGwToTwhPerYear \cdot VCorrBaseLoad(runCy, YTIME))}{(VElecPeakLoad(runCy, YTIME) - VCorrBaseLoad(runCy, YTIME))}$$

- Compute electricity final demand

$$\begin{aligned} VElecDem(runCy, YTIME) = \frac{1}{sTWhToMtoe} \cdot (&VFeCons(runCy, "ELC", YTIME) \\ &+ VFNonEnCons(runCy, "ELC", YTIME) \\ &+ VLosses(runCy, "ELC", YTIME) \\ &+ VEnCons(runCy, "ELC", YTIME) \\ &- VNetImp(runCy, "ELC", YTIME)) \end{aligned}$$

- Compute estimated base load

$$VEstBaseLoad(runCy, YTIME) = \frac{\left(\sum_{DSBS}(iBaseLoadShareDem(runCy,DSBS,YTIME) \cdot VElecConsAll(runCy,DSBS,YTIME)) \cdot (1 + iRateLossesFinCons(runCy,"E\right.}{sT}$$
$$sGwTo$$

- Compute load factor of entire domestic system

$$VCapChpPlants(runCy, YTIME) = \frac{(\sum_{INDDOM} VConsFuel(runCy, INDDOM, "ELC", YTIME) + \sum_{TRANSE} VDemTr(runCy, TRANSE,}{\left(\sum_{INDDOM} \left(\frac{VConsFuel(runCy,INDDOM,"ELC",YTIME)}{iLoadFacElecDem(runCy,INDDOM,YTIME)}\right) + \sum_{TRANSE} \left(\frac{VDemTr(runCy,TRANSE,"ELC",}{iLoadFacElecDem(runCy,TRANSE,}\right.\right.}$$

- Compute elerctricity peak load

$$VElecPeakLoad(runCy, YTIME) = \frac{VElecDem(runCy, YTIME)}{(VCapChpPlants(runCy, YTIME) \cdot sGwToTwhPerYear)}$$

- Compute baseload corresponding to maximum load

$$\begin{aligned}&(VElecDem(runCy, YTIME) - VBslMaxmLoad(runCy, YTIME) \cdot sGwToTwhPerYear)\\ &\quad = iMxmLoadFacElecDem(runCy, YTIME)\\ &\qquad \cdot (VElecPeakLoad(runCy, YTIME) - VBslMaxmLoad(runCy, YTIME))\\ &\qquad \cdot sGwToTwhPerYear\end{aligned}$$

- Compute electricity base load

$$\begin{aligned}&VCorrBaseLoad(runCy, YTIME)\\ &= \left(\frac{1}{(1 + Exp\,(iBslCorrection(runCy, YTIME) \cdot (VEstBaseLoad(runCy, YTIME) - VBslMaxmLoad(run}\right.\\ &\quad \cdot VEstBaseLoad(runCy, YTIME) + \left(1\right.\\ &\quad - \frac{1}{(1 + Exp\,(iBslCorrection(runCy, YTIME) \cdot (VEstBaseLoad(runCy, YTIME) - VBslMaxmLoad(ru}\\ &\quad \cdot VBslMaxmLoad(runCy, YTIME)\end{aligned}$$

- Compute total required electricity production

$$\begin{aligned}VTotReqElecProd(runCy, YTIME) = \sum_{HOUR} &((VElecPeakLoad(runCy, YTIME)\\ &\quad - VCorrBaseLoad(runCy, YTIME))\\ &\quad \cdot exp\,(-VLoadCurveConstr(runCy, YTIME)\\ &\quad \cdot (0.25 + (ord(HOUR) - 1))))\\ &+ 9 \cdot VCorrBaseLoad(runCy, YTIME)\end{aligned}$$

- Compute Estimated total electricity generation capacity

$$\begin{aligned}&VTotElecGenCapEst(runCy, YTIME)\\ &\quad = iResMargTotAvailCap(runCy, "TOT_CAP_RES", YTIME)\\ &\qquad \cdot VTotElecGenCap(runCy, YTIME - 1) \cdot \frac{VElecPeakLoad(runCy, YTIME)}{VElecPeakLoad(runCy, YTIME - 1)}\end{aligned}$$

- Compute total electricity generation capacity

$$VTotElecGenCap(runCy, YTIME) = VTotElecGenCapEst(runCy, YTIME)$$

- Compute hourly production cost used in investment decisions

$$\begin{aligned}
&VHourProdCostTech(runCy, PGALL, HOUR, YTIME) \\
&= \frac{\frac{\left(\left(iDisc(runCy,"PG",YTIME)\cdot\frac{exp(iDisc(runCy,"PG",YTIME)\cdot iTechLftPlaType(runCy,PGALL))}{(exp(iDisc(runCy,"PG",YTIME)\cdot iTechLftPlaType(runCy,PGALL))-1)}\right)\cdot iGrossCapCosSubRen(runCy,PG\right.}{iAvailRate(PGALL,YTIME)}}{(1000\cdot(ord(HOUR)-1+0.25))} \\
&+\frac{iVarCost(PGALL,YTIME)}{1E3}+(VRenValue(YTIME)\cdot 8.6e \\
&-5)\,\$\,(not\,(PGREN(PGALL)\$\,(notsameas("PGASHYD",PGALL))\,\$\,(notsameas("PGSHYD",PGAL \\
&+\sum_{PGEF\$PGALLtoEF(PGALL,PGEF)}\left(\left(VFuelPriceSub(runCy,"PG",PGEF,YTIME)\right.\right. \\
&\quad+iCO2CaptRate(runCy,PGALL,YTIME)\cdot VCO2CO2SeqCsts(runCy,YTIME)\cdot 1e \\
&\quad-3\cdot iCo2EmiFac(runCy,"PG",PGEF,YTIME) \\
&\quad+(1-iCO2CaptRate(runCy,PGALL,YTIME))\cdot 1e-3 \\
&\quad\cdot iCo2EmiFac(runCy,"PG",PGEF,YTIME) \\
&\quad\cdot\left(\sum_{NAP\$NAPtoALLSBS(NAP,"PG")}VCarVal(runCy,NAP,YTIME)\right)\Bigg) \\
&\quad\cdot\frac{sTWhToMtoe}{iPlantEffByType(runCy,PGALL,YTIME)}\Bigg)\,\$\,(notPGREN(PGALL))
\end{aligned}$$

- Compute hourly production cost used in investment decisions excluding CCS

$$\begin{aligned}
&VHourProdCostTechNoCCS(runCy, PGALL, HOUR, YTIME) \\
&\quad= VPowerPlantNewEq(runCy, PGALL, YTIME) \\
&\quad\cdot VHourProdCostTech(runCy, PGALL, HOUR, YTIME) \\
&\quad+\sum_{CCS\$CCS_NOCCS(CCS,PGALL)}(VPowerPlaShrNewEq(runCy, CCS, YTIME) \\
&\quad\cdot VHourProdCostTech(runCy, CCS, HOUR, YTIME))
\end{aligned}$$

- Compute gamma parameter used in CCS/No CCS decision tree

$$\begin{aligned}
&VSensCcs(runCy, YTIME) \\
&= 20+25\cdot EXP\Big(-0.06 \\
&\quad\cdot\Big(\sum(NAP\$NAPtoALLSBS(NAP,"PG"), VCarVal(runCy, NAP, YTIME-1))\Big)\Big)
\end{aligned}$$

- Compute hourly production cost used in investment decisions taking into account CCS acceptance

$$VHourProdCostTechAfterCCS(runCy, PGALL, HOUR, YTIME) = VHourProdCostTech(runCy, PGALL, HOUR, YTIME)^{(-VSensCcs(runCy,YTIME))}$$

- Compute production cost used in investment decisions

$$VProdCostTechnology(runCy, PGALL, YTIME) = \sum_{HOUR} \left(VHourProdCostTech(runCy, PGALL, HOUR, YTIME)^{(-VSensCcs(runCy,YTIME))}\right)$$

- Compute SHRCAP

$$VPowerPlaShrNewEq(runCy, PGALL, YTIME) = 1.1 \cdot \frac{VProdCostTechnology(runCy, PGALL, YTIM}{(1.1 \cdot VProdCostTechnology(runCy, PGALL, YTIME) + \sum (PGALL2\$CCS_NOCCS(PGALL, PGAL}$$

- Compute SHRCAP excluding CCs

$$VPowerPlantNewEq(runCy, PGALL, YTIME) = 1 - \sum_{CCS\$CCS_NOCCS(CCS,PGALL)} VPowerPlaShrNewEq(runCy, CCS, YTIME)$$

- Compute variable cost of technology

$$VVarCostTech(runCy, PGALL, YTIME) = \left(\frac{iVarCost(PGALL, YTIME)}{1E3} + \sum_{PGEF\$PGALLtoEF(PGALL,PGEF)} \left(\left(\frac{VFuelPriceSub(runCy, "PG", PGEF, YTIME)}{1.2441} + iCO2CaptRate(ru \cdot \frac{sTWhToMtoe}{iPlantEffByType(runCy, PGALL, YTIME)}\right) \$ (notPGREN(PGALL))\right)\right.$$

- Compute variable cost of technology excluding PGSCRN

$$VVarCostTechNotPGSCRN(runCy, PGALL, YTIME) = VVarCostTech(runCy, PGALL, YTIME)^{(-5)}$$

- Compute production cost of technology used in premature replacement

$$VProdCostTechPreReplac(runCy, PGALL, YTIME) = \left(\frac{\left(\left(iDisc(runCy, "PG", YTIME) \cdot \frac{exp(iDisc(runCy,"PG",YTIME) \cdot iTechLftPlaType(runCy,PGALL))}{(exp(iDisc(runCy,"PG",YTIME) \cdot iTechLftPlaType(runCy,PGALL))-1)}\right) \cdot iInvCos}{(8760 \cdot iAvailRate(PGAL}\right.$$

$$+ \left(\frac{iVarCost(PGALL, YTIME)}{1E3} + \sum_{PGEF\$PGALLtoEF(PGALL,PGEF)} \left(\left(VFuelPriceSub(runCy, "PG", PGEF, YTIME) + iCO2CaptRate(ru\right.\right.\right.$$

- Compute production cost of technology used in premature replacement including plant availability rate

$$
\begin{aligned}
&VProdCostTechPreReplacAvail(runCy, PGALL, PGALL2, YTIME) \\
&\quad = \frac{iAvailRate(PGALL, YTIME)}{iAvailRate(PGALL2, YTIME)} \cdot VProdCostTechPreReplac(runCy, PGALL, YTIME) \\
&\qquad + VVarCostTech(runCy, PGALL, YTIME) \cdot \left(1 - \frac{iAvailRate(PGALL, YTIME)}{iAvailRate(PGALL2, YTIME)}\right)
\end{aligned}
$$

- Compute endogenous scrapping index

$$
\begin{aligned}
&VEndogScrapIndex(runCy, PGALL, YTIME) \\
&= \frac{VVarCostTechNotPGSCRN(runC}{\Big(VVarCostTechNotPGSCRN(runCy, PGALL, YTIME) + (iScaleEndogScrap(runCy, PGALL, YTIM}
\end{aligned}
$$

- Compute total electricity generation capacity excluding CHP plants

$$
\begin{aligned}
&VElecGenNoChp(runCy, YTIME) \\
&\quad = VTotElecGenCap(runCy, YTIME) \\
&\qquad - SUM\,(CHP, VElecCapChpPla(runCy, CHP, YTIME) \cdot 0.85)
\end{aligned}
$$

- Compute the gap in power generation capacity

$$
\begin{aligned}
&VGapPowerGenCap(runCy, YTIME) \\
&= \frac{\Big(\Big(VElecGenNoChp(runCy, YTIME) - VElecGenNoChp(runCy, YTIME - 1) + \sum_{PGALL} (VElecGenP}{}
\end{aligned}
$$

- Compute temporary variable facilitating the scaling in Weibull equation

$$
\begin{aligned}
&VScalWeibull(runCy, PGALL, HOUR, YTIME) \\
&= (VHourProdCostTech(runCy, PGALL, HOUR, YTIME)\$\,(notNOCCS(PGALL)) \\
&\quad + VHourProdCostTechNoCCS(runCy, PGALL, HOUR, YTIME)\$NOCCS(PGALL))^{(-6)}
\end{aligned}
$$

- Compute renewable potential supply curve

$$
\begin{aligned}
&VRenPotSupplyCurve(runCy, PGRENEF, YTIME) \\
&\quad = iMinRenPotential(runCy, PGRENEF, YTIME) \\
&\qquad + \frac{VCarVal(runCy, \text{"}Trade\text{"}, YTIME)}{70} \\
&\qquad \cdot (iMaxRenPotential(runCy, PGRENEF, YTIME) \\
&\qquad\qquad - iMinRenPotential(runCy, PGRENEF, YTIME))
\end{aligned}
$$

- Compute maximum allowed renewable potential

$$VMaxmAllowRenPotent(runCy, PGRENEF, YTIME) = \frac{(VRenPotSupplyCurve(runCy, PGRENEF, YTIME) + iMaxRenPotential(runCy, PGRENEF, YTIM}{2}$$

- Compute minimum allowed renewable potential

$$VMnmAllowRenPot(runCy, PGRENEF, YTIME) = \frac{(VRenPotSupplyCurve(runCy, PGRENEF, YTIME) + VMnmAllowRenPot(runCy, PGRENEF, YTI}{2}$$

- Compute renewable technologies maturity multiplier

$$VRenTechMatMult(runCy, PGALL, YTIME) = 1\$\,(NOTPGREN(PGALL)) + \left(\frac{1}{\left(1 + EXP\left(9 \cdot \left(\sum_{PGRENEF\$PGALLtoPGRENEF(PGALL,PGRENEF)} \left(\frac{\sum_{PGALL2\$(PGALLtoPGRENEF(PGALL2,PG}}{VRenPo}\right.\right.\right.\right.\right.$$

- Compute temporary variable facilitating the scaling in Weibull equation

$$\begin{aligned} VScalWeibullSum(runCy, PGALL, YTIME) &= iMatFacPlaAvailCap(runCy, PGALL, YTIME) \\ &\cdot VRenTechMatMult(runCy, PGALL, YTIME) \\ &\cdot \sum_{HOUR} \Big((VHourProdCostTech(runCy, PGALL, HOUR, YTIME)\$\,(notNOCCS(PGALL)) \\ &+ VHourProdCostTechNoCCS(runCy, PGALL, HOUR, YTIME)\$NOCCS(PGALL))^{(-6)} \Big) \end{aligned}$$

- Compute for Power Plant new investment decision

$$VNewInvDecis(runCy, YTIME) = \sum_{PGALL\$(notCCS(PGALL))} VScalWeibullSum(runCy, PGALL, YTIME)$$

- Compute the power plant share in new equipment

$$\begin{aligned} VPowPlaShaNewEquip(runCy, PGALL, YTIME) &= \left(\frac{VScalWeibullSum(runCy, PGALL, YTIME)}{VNewInvDecis(runCy, YTIME)} \right) \$\,(notCCS(PGALL)) \\ &+ \sum_{NOCCS\$CCS_NOCCS(PGALL,NOCCS)} VPowPlaShaNewEquip(runCy, NOCCS, YTIME)\$CCS(PGALL) \end{aligned}$$

- Compute electricity generation capacity

$$
\begin{aligned}
&VElecGenPlantsCapac(runCy, PGALL, YTIME) \\
&= (VElecGenPlanCap(runCy, PGALL, YTIME - 1) \\
&\qquad \cdot VEndogScrapIndex(runCy, PGALL, YTIME - 1) \\
&\qquad + (VPowPlaShaNewEquip(runCy, PGALL, YTIME) \\
&\qquad \cdot VGapPowerGenCap(runCy, YTIME))\,\$\,((notCCS(PGALL))\,\&\,(notNOCCS(PGALL))) \\
&\qquad + (VPowPlaShaNewEquip(runCy, PGALL, YTIME) \\
&\qquad \cdot VPowerPlantNewEq(runCy, PGALL, YTIME) \\
&\qquad \cdot VGapPowerGenCap(runCy, YTIME))\,\$NOCCS(PGALL) \\
&\qquad + (VPowPlaShaNewEquip(runCy, PGALL, YTIME) \\
&\qquad \cdot VPowerPlantNewEq(runCy, PGALL, YTIME) \\
&\qquad \cdot VGapPowerGenCap(runCy, YTIME))\,\$CCS(PGALL) \\
&\qquad + iDecInvPlantSched(runCy, PGALL, YTIME) \cdot iAvailRate(PGALL, YTIME) \\
&\qquad - iPlantDecomSched(runCy, PGALL, YTIME) \cdot iAvailRate(PGALL, YTIME)) \\
&\quad - \Bigg((VElecGenPlantsCapac(runCy, PGALL, YTIME - 1) \\
&\qquad - iPlantDecomSched(runCy, PGALL, YTIME - 1)) \cdot iAvailRate(PGALL, YTIME) \\
&\qquad \cdot \left(\frac{1}{iTechLftPlaType(runCy, PGALL)}\right)\Bigg)\,\$PGSCRN(PGALL)
\end{aligned}
$$

- Compute electricity generation capacity

$$
\begin{aligned}
&VElecGenPlanCap(runCy, PGALL, YTIME) \\
&= \frac{(VElecGenPlantsCapac(runCy, PGALL, YTIME) + 1e - 6 + SQRT\,(SQR\,(VElecGenPlantsCapac(run}{2}
\end{aligned}
$$

- Compute variable cost of technology

$$
\begin{aligned}
&VVarCostTechnology(runCy, PGALL, YTIME) \\
&\quad = iMatureFacPlaDisp(runCy, PGALL, YTIME) \\
&\qquad \cdot VVarCostTechnology(runCy, PGALL, YTIME)^{(-2)}
\end{aligned}
$$

- Compute Electricity peak loads

$$
\begin{aligned}
&VElecPeakLoads(runCy, YTIME) \\
&\quad = \sum_{PGALL} VVarCostTechnology(runCy, PGALL, YTIME)
\end{aligned}
$$

- Compute Power plants sorting according to variable cost to decide the plant dispatching

$$
\begin{aligned}
&VPowPlantSorting(runCy, PGALL, YTIME) \\
&\quad = \frac{VVarCostTechnology(runCy, PGALL, YTIME)}{VElecPeakLoads(runCy, YTIME)}
\end{aligned}
$$

- Compute the new capacity added every year

$$\begin{aligned}&VNewCapYearly(runCy, PGALL, YTIME)\\&\quad = VElecGenPlanCap(runCy, PGALL, YTIME)\\&\qquad - VElecGenPlanCap(runCy, PGALL, YTIME - 1)\end{aligned}$$

- Compute the average capacity factor of RES

$$\begin{aligned}&VAvgCapFacRes(runCy, PGALL, YTIME)\\&= \frac{(iAvailRate(PGALL, YTIME) \cdot VNewCapYearly(runCy, PGALL, YTIME) + iAvailRate(PGALL, YT}{}\end{aligned}$$

- Compute overall capacity

$$\begin{aligned}&VOverallCap(runCy, PGALL, YTIME)\\&\quad = VElecGenPlanCap(runCy, pgall, ytime)\$\,(notPGREN(PGALL))\\&\qquad + VAvgCapFacRes(runCy, PGALL, YTIME - 1)\\&\qquad \cdot \left(\frac{VNewCapYearly(runCy, PGALL, YTIME)}{iAvailRate(PGALL, YTIME)}\right.\\&\qquad\qquad \left. + \frac{VOverallCap(runCy, PGALL, YTIME - 1)}{VAvgCapFacRes(runCy, PGALL, YTIME - 1)}\right)\$PGREN(PGALL)\end{aligned}$$

- Compute the scaling factor for plant dispatching

$$\begin{aligned}&\sum_{PGALL}\left(\left(VOverallCap(runCy, PGALL, YTIME)\right.\right.\\&\quad + \sum_{CHP\$CHPtoEON(CHP,PGALL)} VElecCapChpPla(runCy, CHP, YTIME)\Bigg)\\&\quad \cdot exp\left(-\frac{VScalFacPlaDisp(runCy, HOUR, YTIME)}{VPowPlantSorting(runCy, PGALL, YTIME)}\right)\Bigg)\\&\qquad = (VElecPeakLoad(runCy, YTIME) - VCorrBaseLoad(runCy, YTIME))\\&\qquad\quad \cdot exp\left(-VLoadCurveConstr(runCy, YTIME) \cdot (0.25 + ord(HOUR) - 1)\right)\\&\qquad\quad + VCorrBaseLoad(runCy, YTIME)\end{aligned}$$

- Compute estimated electricity of CHP Plants

$$\begin{aligned}&VElecChpPlants(runCy, YTIME)\\&= \frac{\left(\left(\frac{1}{0.086} \cdot \sum_{INDDOM,CHP} VConsFuel(runCy, INDDOM, CHP, YTIME) \cdot VElecIndPrices(runCy, YTIM\right.\right.}{}\end{aligned}$$

- Compute non CHP electricity production

$$\begin{aligned}VNonChpElecProd(runCy, YTIME) = (&VElecDem(runCy, YTIME)\\&- VElecChpPlants(runCy, YTIME))\end{aligned}$$

- Compute total required electricity production

$$VReqElecProd(runCy, YTIME) = \sum_{hour} \left(\sum_{CHP} \left(VElecCapChpPla(runCy, CHP, YTIME) \cdot exp\left(- \frac{VScalFacPlaDisp(runCy, HOUR, YTIME)}{\sum (pgall\$chptoeon(chp, pgall), VPowPlantSorting(runCy, PGALL, YTIME))} \right) \right) \right)$$

$$\begin{aligned} & VElecProdPowGenPlants(runCy, PGALL, YTIME) \\ & = \frac{VNonChpElecProd(runCy, YTIME)}{(VTotReqElecProd(runCy, YTIME) - VReqElecProd(runCy, YTIME))} \\ & \cdot VElecGenPlanCap(runCy, PGALL, YTIME) \\ & \cdot \sum_{HOUR} \left(exp\left(- \frac{VScalFacPlaDisp(runCy, HOUR, YTIME)}{VPowPlantSorting(runCy, PGALL, YTIME)} \right) \right) \end{aligned}$$

$$\begin{aligned} & VSecContrTotChpProd(runCy, INDDOM, CHP, YTIME) \\ & = \frac{VConsFuel(runCy, INDDOM, CHP, YTIME)}{(1e-6 + SUM\,(INDDOM2, VConsFuel(runCy, INDDOM2, CHP, YTIME)))} \end{aligned}$$

$$\begin{aligned} & VChpElecProd(runCy, CHP, YTIME) \\ & = \frac{\sum_{INDDOM} VConsFuel(runCy, INDDOM, CHP, YTIME)}{SUM\,(chp2, \sum_{INDDOM} VConsFuel(runCy, INDDOM, CHP2, YTIME))} \\ & \cdot (VElecDem(runCy, YTIME) \\ & \qquad - SUM\,(PGALL, VElecProd(runCy, PGALL, YTIME))) \end{aligned}$$

$$\begin{aligned} & VResShareGrossElecProd(runCy, YTIME) \\ & = \frac{(SUM\,(PGNREN\$\,((notsameas("PGASHYD", PGNREN))\,\$\,(notsameas("PGSHY}{\left(SUM\,(PGALL, VElecProdPowGenPlants(runCy, PGALL, YTIME)) + 1e-3 \cdot \frac{\sum_{DSBS}\left(\sum_{CHP\$SECTTE}\right.}{} \right.} \end{aligned}$$

$$\begin{aligned} & VLongPowGenCost(runCy, PGALL, ESET, YTIME) \\ & = \frac{\frac{\left(iDisc(runCy,"PG",YTIME) \cdot \frac{EXP(iDisc(runCy,"pg",YTIME) \cdot iTechLftPlaType(runCy,PGALL))}{(EXP(iDisc(runCy,"PG",YTIME) \cdot iTechLftPlaType(runCy,PGALL)) - 1)} \cdot iGrossCapCosSubRen(runCy,PGA\right.}{iAvailRate(PGALL,YTIME)}}{(1000 \cdot (6\$ISET(ESET) + 4\$RSET(ESET} \\ & + \sum_{PGEF\$PGALLTOEF(PGALL,PGEF)} \left(\left(\frac{iVarCost(PGALL, YTIME)}{1000} \right.\right. \\ & + \left(\frac{VFuelPriceSub(runCy, "PG", PGEF, YTIME)}{1.2441} + iCO2CaptRate(runCy, PGALL, YTIME) \cdot VCO2 \right. \\ & \qquad \left.\left.\left. \cdot \frac{sTWhToMtoe}{iPlantEffByType(runCy, PGALL, YTIME)} \right) \right) \end{aligned}$$

$$VLonMnmpowGenCost(runCy, PGALL, YTIME)$$

$$= \frac{\frac{\left(iDisc(runCy,"PG",YTIME)\cdot\frac{EXP(iDisc(runCy,"pg",YTIME)\cdot iTechLftPlaType(runCy,PGALL))}{(EXP(iDisc(runCy,"PG",YTIME)\cdot iTechLftPlaType(runCy,PGALL))-1)}\cdot iGrossCapCosSubRen(runCy,PGA\right.}{iAvailRate(PGALL,YTIME)}}{(1000\cdot sGwToTwhPerYear)}$$

$$+ \sum_{PGEF\$PGALLTOEF(PGALL,PGEF)} \left(\left(\frac{iVarCost(PGALL,YTIME)}{1000}\right.\right.$$

$$+ \left(\frac{VFuelPriceSub(runCy,"PG",PGEF,YTIME)}{1.2441} + iCO2CaptRate(runCy,PGALL,YTIME)\cdot VCO2\right.$$

$$\left.\left.\cdot\frac{sTWhToMtoe}{iPlantEffByType(runCy,PGALL,YTIME)}\right)\right)$$

$$VLongPowGenIntPri(runCy, PGALL, ESET, YTIME)$$

$$= \frac{\frac{\left(iDisc(runCy,"PG",YTIME)\cdot\frac{EXP(iDisc(runCy,"pg",YTIME)\cdot iTechLftPlaType(runCy,PGALL))}{(EXP(iDisc(runCy,"PG",YTIME)\cdot iTechLftPlaType(runCy,PGALL))-1)}\cdot\frac{iGrossCapCosSubRen(runCy,PGALL,}{1.5}\right.}{iAvailRate(PGALL,YTIME)}}{(1000\cdot(7.25\$ISET(ESET)+2.25\$RSET(ESET}$$

$$+ \sum_{PGEF\$PGALLTOEF(PGALL,PGEF)} \left(\left(\frac{iVarCost(PGALL,YTIME)}{1000}\right.\right.$$

$$+ \left(\left(SUM\left(EF, \sum_{WEF\$EFtoWEF("PG",EF,WEF)} iPriceFuelsInt(WEF,YTIME)\cdot\frac{sTWhToMtoe}{1000}\cdot 1.5\right)\right)\$(\right.$$

$$\left.\left.\cdot\frac{sTWhToMtoe}{iPlantEffByType(runCy,PGALL,YTIME)}\right)\right)$$

$$VShoPowGenIntPri(runCy, PGALL, ESET, YTIME)$$

$$= \sum_{PGEF\$PGALLTOEF(PGALL,PGEF)} \left(\left(\frac{iVarCost(PGALL,YTIME)}{1000}\right.\right.$$

$$+ \left(\left(SUM\left(EF, \sum_{WEF\$EFtoWEF("PG",EF,WEF)} iPriceFuelsInt(WEF,YTIME)\cdot\frac{sTWhToMtoe}{1000}\cdot 1.5\right)\right)\$(\right.$$

$$\left.\left.\cdot\frac{sTWhToMtoe}{iPlantEffByType(runCy,PGALL,YTIME)}\right)\right)$$

$$VLongAvgPowGenCost(runCy, ESET, YTIME)$$

$$= \frac{(SUM\,(PGALL, VElecProdPowGenPlants(runCy,PGALL,YTIME)\cdot VLongPowGenCost(runCy,PG}{VElec}$$

$$VLonAvgPowGenCostNoClimPol(runCy, PGALL, ESET, YTIME)$$
$$= \frac{\frac{\left(iDisc(runCy,"PG",YTIME)\cdot\frac{EXP(iDisc(runCy,"pg",YTIME)\cdot iTechLftPlaType(runCy,PGALL))}{(EXP(iDisc(runCy,"PG",YTIME)\cdot iTechLftPlaType(runCy,PGALL))-1)}\cdot iGrossCapCosSubRen(runCy,PGA\right.}{iAvailRate(PGALL,YTIME)}}{(1000\cdot(7.25\$ISET(ESET)+2.25\$RSET(ESI}$$
$$+\sum_{PGEF\$PGALLTOEF(PGALL,PGEF)}\left(\left(\frac{iVarCost(PGALL,YTIME)}{1000}\right.\right.$$
$$+\left(\frac{\left(VFuelPriceSub(runCy,"PG",PGEF,YTIME)-\frac{iEffValueInEuro(runCy,"pg",ytime)}{1000}-iCo2EmiFac(r\right.}{1.2441}\right.$$
$$\left.\left.\left.\cdot\frac{sTWhToMtoe}{iPlantEffByType(runCy,PGALL,YTIME)}\right)\right)\right)$$

$$VLonPowGenCostNoClimPol(runCy, ESET, YTIME)$$
$$= \frac{(SUM\,(PGALL, VElecProdPowGenPlants(runCy, PGALL, YTIME)\cdot VLonAvgPowGenCostNoClimP}{}$$

$$VElecPriIndResNoCliPol(runCy, ESET, YTIME)$$
$$= (1+iFacElecPriConsu(runCy,"VAT",YTIME))$$
$$\cdot(((VFuelPriceSub(runCy,"OI","ELC",YTIME-1)\cdot sTWhToMtoe)\,\$TFIRST(YTIME-1)$$
$$+(iFacElecPriConsu(runCy,"IND_RES",YTIME-1)+VRenShareElecProdSub(runCy,YTIME-1$$
$$+((VFuelPriceSub(runCy,"HOU","ELC",YTIME-1)\cdot sTWhToMtoe)\,\$TFIRST(YTIME-1)$$
$$+(iFacElecPriConsu(runCy,"TERT_RES",YTIME-1)+VRenShareElecProdSub(runCy,YTIME-$$

$$VAvgPowerGenCostShoTrm(runCy, ESET, YTIME)$$
$$= \frac{\left(\sum_{PGALL}\left(VElecProdPowGenPlants(runCy, PGALL, YTIME)\cdot\left(\sum_{PGEF\$PGALLtoEF(PGALL,PGEF)}\left(\left(\frac{iV}{}\right.\right.\right.\right.}{}$$

$$VLifeTimeTech(runCy, DSBS, EF, TEA, YTIME) = \frac{1}{VScrRate(runCy, YTIME)}$$

$$
\begin{aligned}
&VGoodsTranspActiv(runCy, TRANSE, YTIME) \\
&= \Bigg(iResActiv(runCy, TRANSE, YTIME) \\
&\qquad \cdot VGoodsTranspActiv(runCy, TRANSE, YTIME-1) \\
&\qquad \cdot \frac{\left[\left(\frac{iGDP(YTIME,runCy)}{iPop(YTIME,runCy)}\right)\right.}{\left.\left(\frac{iGDP(YTIME-1,runCy)}{iPop(YTIME-1,runCy)}\right)\right]^{iElastA(runCy,TRANSE,"a",YTIME)}} \cdot \left(\frac{iPop(YTIME, runCy)}{iPop(YTIME-1, runCy)}\right) \\
&\qquad \cdot \left(\frac{VFuelPriceAvg(runCy, TRANSE, YTIME)}{VFuelPriceAvg(runCy, TRANSE, YTIME-1)}\right)^{iElastA(runCy,TRANSE,"c1",YTIME)} \\
&\qquad \cdot \left(\frac{VFuelPriceAvg(runCy, TRANSE, YTIME-1)}{VFuelPriceAvg(runCy, TRANSE, YTIME-2)}\right)^{iElastA(runCy,TRANSE,"c2",YTIME)} \\
&\quad \cdot \prod_{kpdl} \left(\frac{\left[\left(\frac{VFuelPriceAvg(runCy,TRANSE,YTIME-ord(kpdl))}{VFuelPriceAvg(runCy,TRANSE,YTIME-(ord(kpdl)+1))}\right)\right.}{\left.\left(iCGI(runCy, YTIME)^{\left(\frac{1}{6}\right)}\right)\right]^{(iElastA(runCy,TRANSE,"c3",YTIME)\cdot iFPDL(TRANSE,KPDL))}} \right) \Bigg) \$sameas(\\
&\quad + \Bigg(iResActiv(runCy, TRANSE, YTIME) \\
&\qquad \cdot VGoodsTranspActiv(runCy, TRANSE, YTIME-1) \\
&\qquad \cdot \frac{\left[\left(\frac{iGDP(YTIME,runCy)}{iPop(YTIME,runCy)}\right)\right.}{\left.\left(\frac{iGDP(YTIME-1,runCy)}{iPop(YTIME-1,runCy)}\right)\right]^{iElastA(runCy,TRANSE,"a",YTIME)}} \\
&\qquad \cdot \left(\frac{VFuelPriceAvg(runCy, TRANSE, YTIME)}{VFuelPriceAvg(runCy, TRANSE, YTIME-1)}\right)^{iElastA(runCy,TRANSE,"c1",YTIME)} \\
&\qquad \cdot \left(\frac{VFuelPriceAvg(runCy, TRANSE, YTIME-1)}{VFuelPriceAvg(runCy, TRANSE, YTIME-2)}\right)^{iElastA(runCy,TRANSE,"c2",YTIME)} \\
&\quad \cdot \prod_{kpdl} \left(\frac{\left[\left(\frac{VFuelPriceAvg(runCy,TRANSE,YTIME-ord(kpdl))}{VFuelPriceAvg(runCy,TRANSE,YTIME-(ord(kpdl)+1))}\right)\right.}{\left.\left(iCGI(runCy, YTIME)^{\left(\frac{1}{6}\right)}\right)\right]^{(iElastA(runCy,TRANSE,"c3",YTIME)\cdot iFPDL(TRANSE,KPDL))}} \right) \\
&\quad \cdot \left(\frac{VGoodsTranspActiv(runCy, "GU", YTIME)}{VGoodsTranspActiv(runCy, "GU", YTIME-1)}\right)^{iElastA(runCy,TRANSE,"c4",YTIME)} \Bigg) \$\,(notsameas(TR
\end{aligned}
$$

$$
\begin{aligned}
&VGapTranspFillNewTech(runCy, TRANSE, YTIME) \\
&= VNewReg(runCy, YTIME)\$sameas(TRANSE, "PC") \\
&\quad + \left(\left([VTrnspActiv(runCy, TRANSE, YTIME) - VTrnspActiv(runCy, TRANSE, YTIME - 1) + \frac{}{\left(\sum\right.} \right. \right. \\
&\quad + \left(\left([VGoodsTranspActiv(runCy, TRANSE, YTIME) - VGoodsTranspActiv(runCy, TRANSE, YTI \right. \right.
\end{aligned}
$$

$$
\begin{aligned}
&VSpecificFuelCons(runCy, TRANSE, TTECH, TEA, EF, YTIME) \\
&= iResSpecificFuelConsCost(runCy, TRANSE, TTECH, EF, YTIME) \\
&\quad \cdot VSpecificFuelCons(runCy, TRANSE, TTECH, TEA, EF, YTIME - 1) \\
&\quad \cdot \prod_{KPDL} \left(\left(\frac{VFuelPriceSub\,(runCy, TRANSE, EF, YTIME - ord(KPDL))}{VFuelPriceSub\,(runCy, TRANSE, EF, YTIME - (ord(KPDL) + 1))} \right)^{(iElastA(runCy,TRANSE,"}
\end{aligned}
$$

$$
\begin{aligned}
&VTranspCostPermeanConsSize(runCy, TRANSE, RCon, TTECH, TEA, YTIME) \\
&= \left(\left(\frac{(iDisc(runCy, TRANSE, YTIME) \cdot exp\,(iDisc(runCy, TRANSE, YTIME) \cdot VLifeTimeTech(runC}{(exp\,(iDisc(runCy, TRANSE, YTIME) \cdot VLifeTimeTech(runCy, TRANSE, TTEC} \right. \right. \\
&\qquad \cdot iCapCostTech(runCy, TRANSE, TTECH, YTIME) \cdot iCGI(runCy, YTIME) \\
&\qquad + iFixOMCostTech(runCy, TRANSE, TTECH, YTIME) \\
&\quad + \left(\left(\sum_{EF\$TTECHtoEF(TTECH,EF)} (VSpecificFuelCons(runCy, TRANSE, TTECH, TEA, EF, YTIME) \cdot V \right. \right. \\
&\quad + \left(\sum_{EF\$(TTECHtoEF(TTECH,EF)\$(notsameas("ELC",EF)))} ((1 - iShareAnnMilePlugInHybrid(runCy, YTIME \right. \\
&\qquad + iVarCostTech(runCy, TRANSE, TTECH, YTIME) \\
&\qquad + \left(\frac{VRenValue(YTIME)}{1000} \right) \$\,(notRENEF(TTECH)) \Bigg) \\
&\qquad \cdot iAnnCons(runCy, TRANSE, "smallest") \\
&\qquad \cdot \left(\frac{iAnnCons(runCy, TRANSE, "largest")}{iAnnCons(runCy, TRANSE, "smallest")} \right)^{\left(\frac{(ord(Rcon)-1)}{iNcon(TRANSE)}\right)} \Bigg)
\end{aligned}
$$

$$VTranspCostPerVeh(runCy, TRANSE, rCon, TTECH, TEA, YTIME)$$
$$= VTranspCostPermeanConsSize(runCy, TRANSE, rCon, TTECH, TEA, YTIME)^{(-4)}$$

$$VTranspCostMatFac(runCy, TRANSE, RCon, TTECH, TEA, YTIME)$$
$$= iMatrFactor(runCy, TRANSE, TTECH, YTIME)$$
$$\cdot VTranspCostPerVeh(runCy, TRANSE, rCon, TTECH, TEA, YTIME)$$

$$VTechSortVarCost(runCy, TRANSE, rCon, YTIME)$$
$$= \sum_{(TTECH,TEA)\$SECTTECH(TRANSE,TTECH)} VTranspCostMatFac(runCy, TRANSE, rCon, TTECH, TEA,$$

$$VTechSortVarCostNewEquip(runCy, TRANSE, TTECH, TEA, YTIME)$$
$$= \frac{iMatrFactor(runCy, TRANSE, TTECH, YTIME)}{iCumDistrFuncConsSize(runCy, TRANSE)}$$
$$\cdot \sum_{Rcon\$(ord(Rcon)leiNcon(TRANSE)+1)} \Big(VTranspCostPerVeh(runCy, TRANSE, RCon, TTECH, TEA, YTIM$$
$$\cdot \frac{iDisFunConSize(runCy, TRANSE, RCon)}{VTechSortVarCost(runCy, TRANSE, RCon, YTIME)} \Big)$$

$$VConsEachTechTransp(runCy, TRANSE, TTECH, EF, TEA, YTIME)$$
$$= VConsEachTechTransp(runCy, TRANSE, TTECH, EF, TEA, YTIME - 1)$$
$$\cdot \Bigg(\Bigg(\Bigg(\frac{(VLifeTimeTech(runCy, TRANSE, TTECH, TEA, YTIME - 1) - 1)}{VLifeTimeTech(runCy, TRANSE, TTECH, TEA, YTIME - 1)} \Bigg)$$
$$\cdot \frac{(iAvgVehCapLoadFac(runCy, TRANSE, "CAP", YTIME - 1) \cdot iAvgVehCapLoadFac(runCy, TRAN}{(iAvgVehCapLoadFac(runCy, TRANSE, "CAP", YTIME) \cdot iAvgVehCapLoadFac(runCy, TRAN}$$
$$+ (1 - VScrRate(runCy, YTIME))\,\$sameas(TRANSE, "PC") \Bigg)$$
$$+ VTechSortVarCostNewEquip(runCy, TRANSE, TTECH, TEA, YTIME)$$
$$\cdot \frac{(VSpecificFuelCons(runCy, TRANSE, TTECH, TEA, EF, YTIME)\$\,(notPLUGIN(TTECH)) + (((1}{}$$
$$\cdot VGapTranspFillNewTech(runCy, TRANSE, YTIME)$$
$$\cdot \Bigg(\Bigg(\frac{(iAvgVehCapLoadFac(runCy, TRANSE, "CAP", YTIME - 1) \cdot iAvgVehCapLoadFac(runCy, TRA}{(iAvgVehCapLoadFac(runCy, TRANSE, "CAP", YTIME) \cdot iAvgVehCapLoadFac(runCy, TRA}$$
$$+ VTrnspActiv(runCy, TRANSE, YTIME)\$sameas(TRANSE, "PC") \Bigg)$$

$$VDemTr(runCy, TRANSE, EF, YTIME)$$
$$= \sum_{(TTECH,TEA)\$(SECTTECH(TRANSE,TTECH)\$TTECHtoEF(TTECH,EF))} VConsEachTechTransp(runCy, TRAN$$

$$VFinEneDemTranspSub(runCy, TRANSE, YTIME) = \sum_{EF} VDemTr(runCy, TRANSE, EF, YTIME)$$

$$VMExtV(runCy, YTIME) = iTransChar(runCy, "RES_MEXTV", YTIME) \cdot VMExtV(runCy, YTIME-1) \cdot \frac{\left[\left(\frac{iGDP(YTIME,runCy)}{iPop(YTIME,runCy)}\right)\right.}{\left.\left(\frac{iGDP(YTIME-1,runCy)}{iPop(YTIME-1,runCy)}\right)\right]^{iElastA(runCy,"PC","a",YTIME)}}$$

$$VMExtF(runCy, YTIME) = iTransChar(runCy, "RES_MEXTF", YTIME) \cdot iSigma(runCy, "S1") \cdot EXP\left(iSigma(runCy, "S2") \cdot EXP\left(iSigma(runCy, "S3") \cdot VLamda(runCy, YTIME)\right)\right) \cdot \frac{VNumVeh(runCy, YTIME-1)}{(iPop(YTIME-1, runCy) \cdot 1000)}$$

$$VNumVeh(runCy, YTIME) = \left(\frac{VNumVeh(runCy, YTIME-1)}{(iPop(YTIME-1, runCy) \cdot 1000)} + VMExtF(runCy, YTIME) + VMExtV(runCy, YTIME)\right) \cdot iPop(YTIME, runCy) \cdot 1000$$

$$VNewReg(runCy, YTIME) = (VMExtF(runCy, YTIME) + VMExtV(runCy, YTIME)) \cdot (iPop(YTIME, runCy) \cdot 1000) - VNumVeh(runCy, YTIME-1) \cdot \left(1 - \frac{iPop(YTIME, runCy)}{iPop(YTIME-1, runCy)}\right) + VScrap(runCy, YTIME)$$

$$
\begin{aligned}
&VTrnspActiv(runCy, TRANSE, YTIME) \\
&= \Bigg(iResActiv(runCy, TRANSE, YTIME) \cdot VTrnspActiv(runCy, TRANSE, YTIME-1) \\
&\quad \cdot \left(\frac{VFuelPriceAvg(runCy, TRANSE, YTIME)}{VFuelPriceAvg(runCy, TRANSE, YTIME-1)} \right)^{iElastA(runCy,TRANSE,"b1",YTIME)} \\
&\quad \cdot \left(\frac{VFuelPriceAvg(runCy, TRANSE, YTIME-1)}{VFuelPriceAvg(runCy, TRANSE, YTIME-2)} \right)^{iElastA(runCy,TRANSE,"b2",YTIME)} \\
&\quad \cdot \frac{\left[\left(\frac{VNumVeh(runCy,YTIME-1)}{(iPop(YTIME-1,runCy)\cdot 1000)}\right)\right.}{\left.\left(\frac{VNumVeh(runCy,YTIME)}{(iPop(YTIME,runCy)\cdot 1000)}\right)\right]^{iElastA(runCy,TRANSE,"b3",YTIME)}} \\
&\quad \cdot \frac{\left[\left(\frac{iGDP(YTIME,runCy)}{iPop(YTIME,runCy)}\right)\right.}{\left.\left(\frac{iGDP(YTIME-1,runCy)}{iPop(YTIME-1,runCy)}\right)\right]^{iElastA(runCy,TRANSE,"b4",YTIME)}} \Bigg) \$sameas(TRANSE, "PC") \\
&+ \Bigg(iResActiv(runCy, TRANSE, YTIME) \cdot VTrnspActiv(runCy, TRANSE, YTIME-1) \\
&\quad \cdot \frac{\left[\left(\frac{iGDP(YTIME,runCy)}{iPop(YTIME,runCy)}\right)\right.}{\left.\left(\frac{iGDP(YTIME-1,runCy)}{iPop(YTIME-1,runCy)}\right)\right]^{iElastA(runCy,TRANSE,"a",YTIME)}} \\
&\quad \cdot \left(\frac{VFuelPriceAvg(runCy, TRANSE, YTIME)}{VFuelPriceAvg(runCy, TRANSE, YTIME-1)} \right)^{iElastA(runCy,TRANSE,"c1",YTIME)} \\
&\quad \cdot \left(\frac{VFuelPriceAvg(runCy, TRANSE, YTIME-1)}{VFuelPriceAvg(runCy, TRANSE, YTIME-2)} \right)^{iElastA(runCy,TRANSE,"c2",YTIME)} \Bigg) \$sameas(TRAN \\
&+ \Bigg(iResActiv(runCy, TRANSE, YTIME) \cdot VTrnspActiv(runCy, TRANSE, YTIME-1) \\
&\quad \cdot \frac{\left[\left(\frac{iGDP(YTIME,runCy)}{iPop(YTIME,runCy)}\right)\right.}{\left.\left(\frac{iGDP(YTIME-1,runCy)}{iPop(YTIME-1,runCy)}\right)\right]^{iElastA(runCy,TRANSE,"a",YTIME)}} \\
&\quad \cdot \left(\frac{VFuelPriceAvg(runCy, TRANSE, YTIME)}{VFuelPriceAvg(runCy, TRANSE, YTIME-1)} \right)^{iElastA(runCy,TRANSE,"c1",YTIME)} \\
&\quad \cdot \left(\frac{VFuelPriceAvg(runCy, TRANSE, YTIME-1)}{VFuelPriceAvg(runCy, TRANSE, YTIME-2)} \right)^{iElastA(runCy,TRANSE,"c2",YTIME)} \\
&\quad \cdot \frac{[(VNumVeh(runCy, YTIME) \cdot VTrnspActiv(runCy, "PC", YTIME))}{(VNumVeh(runCy, YTIME-1) \cdot VTrnspActiv(runCy, "PC", YTIME-1))]^{iElastA(runCy,TRANSE,"c4",}} \\
&\quad \cdot \prod_{kpdl} \left(\frac{\left[\left(\frac{VFuelPriceAvg(runCy,TRANSE,YTIME-ord(kpdl))}{VFuelPriceAvg(runCy,TRANSE,YTIME-(ord(kpdl)+1))}\right)\right.}{\left.\left(iCGI(runCy, YTIME)^{\left(\frac{1}{6}\right)}\right)\right]^{(iElastA(runCy,TRANSE,"c3",YTIME)\cdot iFPDL(TRANSE,KPDL))}} \right) \Bigg) \$\,(NOT\,(
\end{aligned}
$$

$$
VScrap(runCy, YTIME) = VScrRate(runCy, YTIME) \cdot VNumVeh(runCy, YTIME-1)
$$

$$VLamda(runCy, YTIME) = \frac{\left(\frac{\left(\frac{VNumVeh(runCy,YTIME-1)}{(iPop(YTIME-1,runCy)\cdot 1000)}\right)}{iPassCarsMarkSat(runCy)} + 1 - SQRT\left(SQR\left(\frac{\left(\frac{VNumVeh(runCy,YTIME-1)}{(iPop(YTIME-1,runCy)\cdot 1000)}\right)}{iPassCarsMarkSat(runCy)} - 1\right)\right)\right)}{2}$$

$$VScrRate(runCy, YTIME) = \frac{[\left(\frac{iGDP(YTIME,runCy)}{iPop(YTIME,runCy)}\right)}{\left(\frac{iGDP(YTIME-1,runCy)}{iPop(YTIME-1,runCy)}\right)]^{0.5}} \cdot VScrRate(runCy, YTIME-1)$$

$$\begin{aligned} &VElecConsAll(runCy, DSBS, YTIME) \\ &= \sum_{INDDOM\$SAMEAS(INDDOM,DSBS)} VConsFuel(runCy, INDDOM, "ELC", YTIME) \\ &\quad + \sum_{TRANSE\$SAMEAS(TRANSE,DSBS)} VDemTr(runCy, TRANSE, "ELC", YTIME) \end{aligned}$$

$$\begin{aligned} &VElecNonSub(runCy, INDDOM, YTIME) \\ &= [iResNonSubsElecDem(runCy, INDDOM, YTIME) \\ &\qquad \cdot VElecNonSub(runCy, INDDOM, YTIME-1) \\ &\qquad \cdot \left(\frac{iActv(YTIME, runCy, INDDOM)}{iActv(YTIME-1, runCy, INDDOM)}\right)^{iElastNonSubElec(runCy,INDDOM,"a",YTIME)} \\ &\quad \cdot \left(\frac{VFuelPriceSub(runCy, INDDOM, "ELC", YTIME)}{VFuelPriceSub(runCy, INDDOM, "ELC", YTIME-1)}\right)^{iElastNonSubElec(runCy,INDDOM,"b1",YTIME)} \\ &\quad \cdot \left(\frac{VFuelPriceSub(runCy, INDDOM, "ELC", YTIME-1)}{VFuelPriceSub(runCy, INDDOM, "ELC", YTIME-2)}\right)^{iElastNonSubElec(runCy,INDDOM,"b2",YTIME)} \\ &\quad \cdot \prod_{KPDL} \left(\left(\frac{VFuelPriceSub\,(runCy, INDDOM, "ELC", YTIME - ord(KPDL))}{VFuelPriceSub\,(runCy, INDDOM, "ELC", YTIME - (ord(KPDL)+1))}\right)^{(iElastNonSubElec(ru} \right. \\ &\qquad \left. - 1, runCy, INDDOM\right) + 0 \end{aligned}$$

$$
\begin{aligned}
&VConsRemSubEquip(runCy, DSBS, EF, YTIME)\\
&= [iResFuelConsPerSubAndFuel(runCy, DSBS, EF, YTIME)\\
&\qquad \cdot \left(\frac{\sum_{TEA}(VLifeTimeTech(runCy, DSBS, EF, TEA, YTIME) - 1)}{\sum_{TEA} VLifeTimeTech(runCy, DSBS, EF, TEA, YTIME)}\right)\\
&\qquad \cdot (VFuelConsInclHP(runCy, DSBS, EF, YTIME - 1)\\
&\quad -(VElecNonSub(runCy, DSBS, YTIME-1)\$\,(ELCEF(EF)\$INDDOM(DSBS))+0\$\,(not\,(ELCEF(E\\
&\qquad \cdot \left(\frac{iActv(YTIME, runCy, DSBS)}{iActv(YTIME - 1, runCy, DSBS)}\right)^{iElastA(runCy,DSBS,"a",YTIME)}\\
&\qquad \cdot \left(\frac{VFuelPriceSub(runCy, DSBS, EF, YTIME)}{VFuelPriceSub(runCy, DSBS, EF, YTIME - 1)}\right)^{iElastA(runCy,DSBS,"b1",YTIME)}\\
&\qquad \cdot \left(\frac{VFuelPriceSub(runCy, DSBS, EF, YTIME - 1)}{VFuelPriceSub(runCy, DSBS, EF, YTIME - 2)}\right)^{iElastA(runCy,DSBS,"b2",YTIME)}\\
&\quad \cdot \prod_{KPDL}\left(\left(\frac{VFuelPriceSub\,(runCy, DSBS, EF, YTIME - ord(KPDL))}{VFuelPriceSub\,(runCy, DSBS, EF, YTIME - (ord(KPDL) + 1))}\right)^{(iElastA(runCy,DSBS,"c",YTIM}\right.\\
&\qquad\qquad - 1, runCy, DSBS) + 0
\end{aligned}
$$

$$
\begin{aligned}
&VDemSub(runCy, DSBS, YTIME) = [VDemSub(runCy, DSBS, YTIME - 1)\\
&\qquad \cdot \left(\frac{iActv(YTIME, runCy, DSBS)}{iActv(YTIME - 1, runCy, DSBS)}\right)^{iElastA(runCy,DSBS,"a",YTIME)}\\
&\qquad \cdot \left(\frac{VFuelPriceAvg(runCy, DSBS, YTIME)}{VFuelPriceAvg(runCy, DSBS, YTIME - 1)}\right)^{iElastA(runCy,DSBS,"b1",YTIME)}\\
&\qquad \cdot \left(\frac{VFuelPriceAvg(runCy, DSBS, YTIME - 1)}{VFuelPriceAvg(runCy, DSBS, YTIME - 2)}\right)^{iElastA(runCy,DSBS,"b2",YTIME)}\\
&\quad \cdot \prod_{KPDL}\left(\left(\frac{\left(\frac{VFuelPriceAvg(runCy,DSBS,YTIME-ord(KPDL))}{VFuelPriceAvg(runCy,DSBS,YTIME-(ord(KPDL)+1))}\right)}{\left(iCGI(runCy, YTIME)^{\left(\frac{1}{6}\right)}\right)}\right)^{(iElastA(runCy,DSBS,"c",YTIME)\cdot iFPDL(DSBS,K}\right.\\
&\qquad\qquad - 1, runCy, DSBS)
\end{aligned}
$$

$$
\begin{aligned}
&VElecConsInd(runCy, YTIME)\\
&\quad = SUM\,(INDSE, VElecNonSub(runCy, INDSE, YTIME))
\end{aligned}
$$

$$
VDemInd(runCy, YTIME) = SUM\,(INDSE, VDemSub(runCy, INDSE, YTIME))
$$

$$
\begin{aligned}
&VElecIndPrices(runCy, YTIME)\\
&= \frac{(VElecIndPricesEst(runCy, YTIME) + sElecToSteRatioChp - SQRT\,(SQR\,(VElecIndPricesEst(run}{2}
\end{aligned}
$$

$$VElecConsHeatPla(runCy, INDDOM, YTIME) = 1E-7$$

$$\begin{aligned}
&VConsFuel(runCy, DSBS, EF, YTIME) \\
&\quad = VConsFuelSub(runCy, DSBS, EF, YTIME)\$\,(notELCEF(EF)) \\
&\qquad + (VConsFuelSub(runCy, DSBS, EF, YTIME) \\
&\qquad\qquad + VElecConsHeatPla(runCy, DSBS, YTIME))\,\$ELCEF(EF)
\end{aligned}$$

$$\begin{aligned}
VElecIndPricesEst(runCy, YTIME) = {} & iResElecIndex(runCy, YTIME) \\
& \cdot VElecIndPrices(runCy, YTIME-1) \\
& \cdot \left(\frac{\left(\frac{VFuelPriceSub(runCy,"OI","ELC",YTIME-1)}{VFuelPriceAvg(runCy,"OI",YTIME-1)}\right)}{\left(\frac{VFuelPriceSub(runCy,"OI","ELC",YTIME-2)}{VFuelPriceAvg(runCy,"OI",YTIME-2)}\right)} \right)^{0.3} \\
& \cdot \left(\frac{\left(\frac{VFuelPriceSub(runCy,"OI","ELC",YTIME-2)}{VFuelPriceAvg(runCy,"OI",YTIME-2)}\right)}{\left(\frac{VFuelPriceSub(runCy,"OI","ELC",YTIME-3)}{VFuelPriceAvg(runCy,"OI",YTIME-3)}\right)} \right)^{0.3}
\end{aligned}$$

$$\begin{aligned}
&VFuePriSubChp(runCy, DSBS, EF, TEA, YTIME) \\
&= \frac{\left(\left(\frac{\left(VFuelPriceSub(runCy,DSBS,EF,YTIME)+\left(\frac{VRenValue(YTIME)}{1000}\right)\$(notRENEF(EF))+\frac{VTechCostVar(runCy,DSBS,EF,TEA,YT}{1000}}{iUsfEneConvSubTech(runCy,DSBS,EF,YTIME)}\right.\right.\right.}{}
\end{aligned}$$

$$\begin{aligned}
&VElecProdCostChp(runCy, DSBS, CHP, YTIME) \\
&= \frac{\frac{\left(\left(iDisc(runCy,"PG",YTIME)\cdot\frac{exp(iDisc(runCy,"PG",YTIME)\cdot iLifChpPla(runCy,DSBS,CHP))}{(exp(iDisc(runCy,"PG",YTIME)\cdot iLifChpPla(runCy,DSBS,CHP))-1)}\right)\cdot iInvCostChp(runCy,DSBS,CHP,Y}{(iAvailRateChp(runCy,DSBS,CHP)\cdot(1000\cdot sTWhT}}{1000} \\
&\quad + \frac{iVarCostChp(runCy, DSBS, CHP, YTIME)}{1000} \\
&\quad + \sum_{PGEF\$CHPtoEF(CHP,PGEF)} \left(\left(VFuelPriceSub(runCy, "PG", PGEF, YTIME) + 0.001 \right.\right. \\
&\qquad\qquad \cdot iCo2EmiFac(runCy, "PG", PGEF, YTIME) \\
&\qquad\qquad \left.\cdot \left(\sum_{NAP\$NAPtoALLSBS(NAP,"PG")} VCarVal(runCy, NAP, YTIME) \right) \right) \\
&\qquad \left. \cdot \frac{sTWhToMtoe}{(iBoiEffChp(runCy, CHP, YTIME) \cdot VElecIndPrices(runCy, YTIME))} \right)
\end{aligned}$$

$$\begin{aligned}
&VTechCost(runCy, DSBS, rCon, EF, TEA, YTIME) \\
&\quad = VTechCostIntrm(runCy, DSBS, rCon, EF, TEA, YTIME)^{(-iElaSub(runCy,DSBS))}
\end{aligned}$$

$$
\begin{aligned}
&VTechCostIntrm(runCy, DSBS, rCon, EF, TEA, YTIME) \\
&= \Bigg(\Bigg(\frac{((iDisc(runCy, DSBS, YTIME)\$\,(notCHP(EF)) + iDisc(runCy, "PG", YTIME)\$CHP(EF)) \cdot exp\,(}{(exp\,((iDisc(runCy, DSBS, YTIME)\$\,(notCHP(EF))} \\
&\qquad \cdot iCapCostTech(runCy, DSBS, EF, YTIME) \cdot iCGI(runCy, YTIME) \\
&\qquad + \frac{iFixOMCostTech(runCy, DSBS, EF, YTIME)}{1000} \\
&\qquad + VFuePriSubChp(runCy, DSBS, EF, TEA, YTIME) \\
&\qquad \cdot iAnnCons(runCy, DSBS, "smallest") \\
&\qquad \cdot \left(\frac{iAnnCons(runCy, DSBS, "largest")}{iAnnCons(runCy, DSBS, "smallest")} \right)^{\left(\frac{(ord(rCon)-1)}{iNcon(DSBS)}\right)} \Bigg) \$INDDOM(DSBS) \\
&+ \Bigg(\Bigg(\frac{(iDisc(runCy, DSBS, YTIME) \cdot exp\,(iDisc(runCy, DSBS, YTIME) \cdot VLifeTimeTech(runCy, DS}{(exp\,(iDisc(runCy, DSBS, YTIME) \cdot VLifeTimeTech(runCy, DSBS, EF, TEA, YT} \\
&\qquad \cdot iCapCostTech(runCy, DSBS, EF, YTIME) \cdot iCGI(runCy, YTIME) \\
&\qquad + \frac{iFixOMCostTech(runCy, DSBS, EF, YTIME)}{1000} \\
&+ \left(\frac{\left(VFuelPriceSub(runCy, DSBS, EF, YTIME) + \frac{VTechCostVar(runCy, DSBS, EF, TEA, YTIME)}{1000}\right)}{iUsfEneConvSubTech(runCy, DSBS, EF, YTIME)} \right) \\
&\qquad \cdot iAnnCons(runCy, DSBS, "smallest") \\
&\qquad \cdot \left(\frac{iAnnCons(runCy, DSBS, "largest")}{iAnnCons(runCy, DSBS, "smallest")} \right)^{\left(\frac{(ord(rCon)-1)}{iNcon(DSBS)}\right)} \Bigg) \$NENSE(DSBS)
\end{aligned}
$$

$$
\begin{aligned}
&VTechCostMatr(runCy, DSBS, rCon, EF, TEA, YTIME) \\
&\quad = iMatrFactor(runCy, DSBS, EF, YTIME) \\
&\qquad \cdot VTechCost(runCy, DSBS, rCon, EF, TEA, YTIME)
\end{aligned}
$$

$$
\begin{aligned}
&VTechSort(runCy, DSBS, rCon, YTIME) \\
&\quad = \sum_{(EF,TEA)\$SECTTECH(DSBS,EF)} VTechCostMatr(runCy, DSBS, rCon, EF, TEA, YTIME)
\end{aligned}
$$

$$
\begin{aligned}
&VGapFinalDem(runCy, DSBS, YTIME) = VDemSub(runCy, DSBS, YTIME) \\
&\quad - \sum_{EF\$SECTTECH(DSBS,EF)} VConsRemSubEquip(runCy, DSBS, EF, YTIME) \\
&\quad + \frac{SQRT\left(SQR\left(VDemSub(runCy, DSBS, YTIME) - \sum_{EF\$SECTTECH(DSBS,EF)} VConsRemSubEquip(r\right.\right.}{2}
\end{aligned}
$$

$$
\begin{aligned}
&VTechShareNewEquip(runCy, DSBS, EF, TEA, YTIME) \\
&\quad = \frac{iMatrFactor(runCy, DSBS, EF, YTIME)}{iCumDistrFuncConsSize(runCy, DSBS)} \\
&\qquad \cdot \sum_{rCon\$(ord(rCon)leiNcon(DSBS)+1)} \Big(VTechCost(runCy, DSBS, rCon, EF, TEA, YTIME) \\
&\qquad\qquad \cdot \frac{iDisFunConSize(runCy, DSBS, rCon)}{VTechSort(runCy, DSBS, rCon, YTIME)} \Big)
\end{aligned}
$$

$$
\begin{aligned}
&VFuelConsInclHP(runCy, DSBS, EF, YTIME) \\
&\quad = VConsRemSubEquip(runCy, DSBS, EF, YTIME) \\
&\qquad + \sum_{TEA} (VTechShareNewEquip(runCy, DSBS, EF, TEA, YTIME) \\
&\qquad\qquad \cdot VGapFinalDem(runCy, DSBS, YTIME)) \\
&\qquad + VElecNonSub(runCy, DSBS, YTIME)\$\,(INDDOM(DSBS)\$ELCEF(EF))
\end{aligned}
$$

$$
\begin{aligned}
&VProCostCHPDem(runCy, DSBS, CHP, YTIME) \\
&\quad = \frac{iVarCostChp(runCy, DSBS, CHP, YTIME)}{1E3} \\
&\qquad + \sum_{PGEF\$CHPtoEF(CHP,PGEF)} \Bigg(\Bigg(VFuelPriceSub(runCy, "PG", PGEF, YTIME) + 1e \\
&\qquad\qquad - 3 \cdot iCo2EmiFac(runCy, "PG", PGEF, YTIME) \\
&\qquad\qquad \cdot \Bigg(\sum_{NAP\$NAPtoALLSBS(NAP,"PG")} VCarVal(runCy, NAP, YTIME) \Bigg) \Bigg) \\
&\qquad\qquad \cdot \frac{sTWhToMtoe}{(iBoiEffChp(runCy, CHP, YTIME) \cdot VElecIndPrices(runCy, YTIME))} \Bigg)
\end{aligned}
$$

$$
\begin{aligned}
&VAvgElcProCHP(runCy, CHP, YTIME) \\
&= \Bigg(\sum_{INDDOM} \Bigg(\frac{VConsFuel(runCy, INDDOM, CHP, YTIME-1)}{SUM\,(INDDOM2, VConsFuel(runCy, INDDOM2, CHP, YTIME-1))} \\
&\quad \cdot VElecProdCostChp(runCy, INDDOM, CHP, YTIME) \Bigg) \Bigg) \$SUM\,(INDDOM2, VConsFuel.L(runCy, \\
&\qquad\qquad - 1)) \\
&\quad + 0\$\,(NOTSUM\,(INDDOM2, VConsFuel.L(runCy, INDDOM2, CHP, YTIME-1)))
\end{aligned}
$$

$$
\begin{aligned}
&VAvgVarProdCostCHP(runCy, CHP, YTIME) \\
&= \left(\sum_{INDDOM} \left(\frac{VConsFuel(runCy, INDDOM, CHP, YTIME-1)}{SUM\,(INDDOM2, VConsFuel(runCy, INDDOM2, CHP, YTIME-1))} \right.\right. \\
&\quad \left.\left. \cdot VProCostCHPDem(runCy, INDDOM, CHP, YTIME) \right) \right) \$SUM\,(INDDOM2, VConsFuel.L(runCy \\
&\qquad\qquad\qquad

$$VTransfInputDHPlants(runCy, EFS, YTIME)$$
$$= \sum_{DH\$DHtoEF(DH,EFS)} \left(\frac{\sum (DOMSE\$SECTTECH(DOMSE, DH), VConsFuel(runCy, DOMSE, DH, YT}{iEffDHPlants(runCy, EFS, YTIME)} \right.$$

$$VTransfInputPatFuel(runCy, EFS, YTIME)$$
$$= \sum_{EF\$(EFS(EF)\$iAvgEffGas(runCy,EF,YTIME))} \left(\frac{VTransfOutputPatFuel(runCy, EFS, YTIME)}{iAvgEffGas(runCy, EF, YTIME)} \right)$$
$$\cdot iShareFueTransfInput(runCy, EFS)$$

$$VRefCapacity(runCy, YTIME) = [iResRefCapacity(runCy, YTIME)$$
$$\cdot VRefCapacity(runCy, YTIME - 1) \cdot \left(1\$ \, (ord(YTIME) le 16) \right.$$
$$+ \left(\prod_{rc} \left(\left(\frac{\sum_{EFS\$EFtoEFA(EFS,"LQD")} (VFeCons\,(runCy, EFS, YTIME - (ord(rc) + 1)))}{\sum_{EFS\$EFtoEFA(EFS,"LQD")} (VFeCons\,(runCy, EFS, YTIME - (ord(rc) + 2)))} \right)^{\left(\frac{0.5}{(ord(rc)+1)}\right)} \right) \right)$$

$$VTransfOutputRefineries(runCy, EFS, YTIME)$$
$$= [iResTransfOutputRefineries(runCy, EFS, YTIME)$$
$$\cdot VTransfOutputRefineries(runCy, EFS, YTIME - 1)$$
$$\cdot \left(\frac{VRefCapacity(runCy, YTIME)}{VRefCapacity(runCy, YTIME - 1)} \right)^{0.3}$$
$$\cdot \left(1\$ \, (TFIRST(YTIME - 1) or TFIRST(YTIME - 2)) \right.$$
$$+ \left(\frac{\sum_{EF\$EFtoEFA(EF,"LQD")} VFeCons(runCy, EF, YTIME - 1)}{\sum_{EF\$EFtoEFA(EF,"LQD")} VFeCons(runCy, EF, YTIME - 2)} \right) \$ \, (not \, (TFIRST(YTIME - 1) or TFIR$$

$$VTransfInputRefineries(runCy, "CRO", YTIME)$$
$$= [VTransfInputRefineries(runCy, "CRO", YTIME - 1)$$
$$\cdot \frac{\sum_{EFS\$EFtoEFA(EFS,"LQD")} VTransfOutputRefineries(runCy, EFS, YTIME)}{\sum_{EFS\$EFtoEFA(EFS,"LQD")} VTransfOutputRefineries(runCy, EFS, YTIME - 1)]\$iRefCapacity\,(runC}$$

$$VTransfOutputNuclear(runCy, "ELC", YTIME)$$
$$= SUM\,(PGNUCL, VElecProd(runCy, PGNUCL, YTIME)) \cdot sTWhToMtoe$$

$$
\begin{aligned}
&VTransfInNuclear(runCy, "NUC", YTIME) \\
&\quad = SUM\left(\frac{PGNUCL, VElecProd(runCy, PGNUCL, YTIME)}{iPlantEffByType(runCy, PGNUCL, YTIME)}\right) \cdot sTWhToMtoe
\end{aligned}
$$

$$
\begin{aligned}
&VTransfInThermPowPls(runCy, PGEF, YTIME) \\
&= \sum_{PGALL\$(PGALLtoEF(PGALL,PGEF)\$((notPGGEO(PGALL))\$(notPGNUCL(PGALL))))} \Bigg(VElecProd(runCy, PGALL, Y \\
&\qquad \cdot \frac{sTWhToMtoe}{iPlantEffByType(runCy, PGALL, YTIME)}\Bigg) \\
&\quad + \sum_{PGALL\$(PGALLtoEF(PGALL,PGEF)\$PGGEO(PGALL))} (VElecProd(runCy, PGALL, YTIME) \\
&\qquad \cdot sTWhToMtoe) \\
&\quad + \frac{\sum_{CHP\$CHPtoEF(CHP,PGEF)} \left(\sum_{INDDOM} VConsFuel(runCy, INDDOM, CHP, YTIME) + sTWhToMt\right.}{\left(0.8 + 0.1 \cdot \frac{(ord(YTIME)-16)}{32}\right)}
\end{aligned}
$$

$$
\begin{aligned}
&VTransfOutThermPowSta(runCy, TOCTEF, YTIME) \\
&= \Bigg(\sum_{PGALL\$(notPGNUCL(PGALL))} VElecProd(runCy, PGALL, YTIME) \cdot sTWhToMtoe \\
&\qquad + \sum_{CHP} (VChpElecProd(runCy, CHP, YTIME) \cdot sTWhToMtoe)\Bigg) \$ELCEF(TOCTEF) \\
&\quad + \Bigg(\sum_{INDDOM} \Bigg(\sum_{CHP\$SECTTECH(INDDOM,CHP)} VConsFuel(runCy, INDDOM, CHP, YTIME)\Bigg) \\
&\qquad + iRateEneBranCons(runCy, TOCTEF, YTIME) \\
&\qquad \cdot (VFeCons(runCy, TOCTEF, YTIME) + VFNonEnCons(runCy, TOCTEF, YTIME) \\
&\qquad + VLosses(runCy, TOCTEF, YTIME)) \\
&\qquad + VLosses(runCy, TOCTEF, YTIME)\Bigg) \$STEAM(TOCTEF)
\end{aligned}
$$

$$
\begin{aligned}
&VTotTransfInput(runCy, EFS, YTIME) \\
&\quad = (VTransfInThermPowPls(runCy, EFS, YTIME) \\
&\qquad + VTransfInputDHPlants(runCy, EFS, YTIME) \\
&\qquad + VTransfInNuclear(runCy, EFS, YTIME) \\
&\qquad + VTransfInputPatFuel(runCy, EFS, YTIME) \\
&\qquad + VTransfInputRefineries(runCy, EFS, YTIME))\,\$\,(notsameas(EFS, "OGS")) \\
&\quad + (VTotTransfOutput(runCy, EFS, YTIME) - VFeCons(runCy, EFS, YTIME) \\
&\qquad - VFNonEnCons(runCy, EFS, YTIME) - iRateEneBranCons(runCy, EFS, YTIME) \\
&\qquad \cdot VTotTransfOutput(runCy, EFS, YTIME) \\
&\qquad - VLosses(runCy, EFS, YTIME))\,\$sameas(EFS, "OGS")
\end{aligned}
$$

$$
\begin{aligned}
&VTotTransfOutput(runCy, EFS, YTIME) \\
&\quad = VTransfOutThermPowSta(runCy, EFS, YTIME) \\
&\qquad + VTransfOutputDHPlants(runCy, EFS, YTIME) \\
&\qquad + VTransfOutputNuclear(runCy, EFS, YTIME) \\
&\qquad + VTransfOutputPatFuel(runCy, EFS, YTIME) \\
&\qquad + VTransfOutputRefineries(runCy, EFS, YTIME)
\end{aligned}
$$

$$
\begin{aligned}
&VTransfers(runCy, EFS, YTIME) \\
&= \Bigg(\Bigg(\Bigg(VTransfers(runCy, EFS, YTIME-1) \cdot iResFeedTransfr(runCy, YTIME) \cdot \frac{VFeCons(runCy, E}{VFeCons(runCy, EFS} \\
&\quad + \Bigg(VTransfers(runCy, "CRO", YTIME-1) \cdot iResFeedTransfr(runCy, YTIME) \cdot \frac{SUM\,(EFS2\$EFT}{SUM\,(EFS2\$EFTC}
\end{aligned}
$$

$$
\begin{aligned}
&VGrsInlConsNotEneBranch(runCy, EFS, YTIME) \\
&\quad = VFeCons(runCy, EFS, YTIME) + VFNonEnCons(runCy, EFS, YTIME) \\
&\qquad + VTotTransfInput(runCy, EFS, YTIME) \\
&\qquad - VTotTransfOutput(runCy, EFS, YTIME) \\
&\qquad + VLosses(runCy, EFS, YTIME) - VTransfers(runCy, EFS, YTIME)
\end{aligned}
$$

$$
\begin{aligned}
VGrssInCons(runCy, EFS, YTIME) = {} & VFeCons(runCy, EFS, YTIME) \\
& + VEnCons(runCy, EFS, YTIME) \\
& + VFNonEnCons(runCy, EFS, YTIME) \\
& + VTotTransfInput(runCy, EFS, YTIME) \\
& - VTotTransfOutput(runCy, EFS, YTIME) \\
& + VLosses(runCy, EFS, YTIME) \\
& - VTransfers(runCy, EFS, YTIME)
\end{aligned}
$$

$$
\begin{aligned}
&VPrimProd(runCy, PPRODEF, YTIME) \\
&= [(iRatePriProTotPriNeeds(runCy, PPRODEF, YTIME) \\
&\quad \cdot (VGrsInlConsNotEneBranch(runCy, PPRODEF, YTIME) + VEnCons(runCy, PPRODEF, YTIME \\
&\qquad + \Bigg(iResHcNgOilPrProd(runCy, PPRODEF, YTIME) \\
&\qquad \cdot VPrimProd(runCy, PPRODEF, YTIME - 1) \\
&\quad \cdot \left(\frac{VGrsInlConsNotEneBranch(runCy, PPRODEF, YTIME)}{VGrsInlConsNotEneBranch(runCy, PPRODEF, YTIME - 1)} \right)^{iNatGasPriProElst(runCy)} \Bigg) \$sameas(P \\
&\qquad + \Bigg(iRatePriProTotPriNeeds(runCy, PPRODEF, YTIME) \\
&\qquad \cdot VPrimProd(runCy, PPRODEF, YTIME - 1) \\
&\quad \cdot \left(\frac{(VGrsInlConsNotEneBranch(runCy, PPRODEF, YTIME) + VExportsFake(runCy, PPRODE}{(VGrsInlConsNotEneBranch(runCy, PPRODEF, YTIME - 1) + VExportsFake(runCy, PPRODE} \right. \\
&\qquad + \Bigg(iResHcNgOilPrProd(runCy, PPRODEF, YTIME) \\
&\qquad \cdot iFuelPriPro(runCy, PPRODEF, YTIME) \\
&\quad \cdot \prod_{kpdl\$(ord(kpdl)lt5)} \left(\left(\frac{iPriceFuelsInt\,(\text{"}WCRO\text{"}, YTIME - (ord(kpdl) + 1))}{iPriceFuelsIntBase\,(\text{"}WCRO\text{"}, YTIME - (ord(kpdl) + 1))} \right)^{(0.2 \cdot iPolDstrbtnLagCoeffPri} \right.
\end{aligned}
$$

$$
\begin{aligned}
VExportsFake(runCy, EFS, YTIME) = &(iFuelExprts(runCy, EFS, YTIME)) \\
&+ iFuelExprts(runCy, EFS, YTIME)
\end{aligned}
$$

$$
\begin{aligned}
&VFkImpAllFuelsNotNatGas(runCy, EFS, YTIME) \\
&= (iRatioImpFinElecDem(runCy, YTIME) \\
&\qquad \cdot (VFeCons(runCy, EFS, YTIME) + VFNonEnCons(runCy, EFS, YTIME)) \\
&\quad + VExportsFake(runCy, EFS, YTIME) + iElecImp(runCy, YTIME))\,\$ELCEF(EFS) \\
&\quad + (VGrssInCons(runCy, EFS, YTIME) + VExportsFake(runCy, EFS, YTIME) \\
&\qquad + VConsFuel(runCy, \text{"}BU\text{"}, EFS, YTIME)\$SECTTECH(\text{"}BU\text{"}, EFS) \\
&\qquad - VPrimProd(runCy, EFS, YTIME))\,\$sameas(EFS, \text{"}CRO\text{"}) \\
&\quad + (VGrssInCons(runCy, EFS, YTIME) + VExportsFake(runCy, EFS, YTIME) \\
&\qquad + VConsFuel(runCy, \text{"}BU\text{"}, EFS, YTIME)\$SECTTECH(\text{"}BU\text{"}, EFS) \\
&\qquad - VPrimProd(runCy, EFS, YTIME))\,\$sameas(EFS, \text{"}NGS\text{"}) \\
&\quad + iImpExp(runCy, \text{"}NGS\text{"}, YTIME)\$sameas(EFS, \text{"}NGS\text{"}) \\
&\quad + ((1 - iRatePriProTotPriNeeds(runCy, EFS, YTIME)) \\
&\qquad \cdot (VGrssInCons(runCy, EFS, YTIME) + VExportsFake(runCy, EFS, YTIME) \\
&\quad + VConsFuel(runCy, \text{"}BU\text{"}, EFS, YTIME)\$SECTTECH(\text{"}BU\text{"}, EFS)))\,\$\,(not\,(ELCEF(EFS) orsame
\end{aligned}
$$

$$
\begin{aligned}
&VNetImports(runCy, EFS, YTIME) \\
&\quad = VFkImpAllFuelsNotNatGas(runCy, EFS, YTIME) \\
&\qquad - VExportsFake(runCy, EFS, YTIME)
\end{aligned}
$$

$$
\begin{aligned}
&VEnCons(runCy, EFS, YTIME) \\
&= iRateEneBranCons(runCy, EFS, YTIME) \cdot ((VTotTransfOutput(runCy, EFS, YTIME) \\
&\quad + VPrimProd(runCy, EFS, YTIME) \$ (sameas(EFS, "CRO") or sameas(EFS, "NGS"))) \$ (notTOCTE \\
&\qquad + (VFeCons(runCy, EFS, YTIME) + VFNonEnCons(runCy, EFS, YTIME) \\
&\qquad\qquad + VLosses(runCy, EFS, YTIME)) \$ TOCTEF(EFS))
\end{aligned}
$$

$$
\begin{aligned}
&VCO2ElcHrgProd(runCy, YTIME) \\
&= \sum_{PGEF} \Bigg(\sum \Bigg(CCS\$PGALLtoEF(CCS, PGEF), VElecProd(runCy, CCS, YTIME) \\
&\qquad \cdot \frac{sTWhToMtoe}{iPlantEffByType(runCy, CCS, YTIME)} \\
&\qquad \cdot iCo2EmiFac(runCy, "PG", PGEF, YTIME) \\
&\qquad \cdot iCO2CaptRate(runCy, CCS, YTIME) \Bigg) \Bigg)
\end{aligned}
$$

$$
\begin{aligned}
VCumCO2Capt(runCy, YTIME) &= VCumCO2Capt(runCy, YTIME - 1) \\
&\quad + VCO2ElcHrgProd(runCy, YTIME - 1)
\end{aligned}
$$

$$
\begin{aligned}
&VWghtTrnstLnrToExpo(runCy, YTIME) \\
&= \frac{1}{\left(1 + exp\left(-iElastCO2Seq(runCy, "mc_s") \cdot \left(\frac{VCumCO2Capt(runCy, YTIME)}{iElastCO2Seq(runCy, "pot")} - iElastCO2Seq(runCy, "mc_m \right.\right.\right.}
\end{aligned}
$$

$$
\begin{aligned}
VCO2CO2SeqCsts(runCy, YTIME) &= (1 - VWghtTrnstLnrToExpo(runCy, YTIME)) \\
&\quad \cdot (iElastCO2Seq(runCy, "mc_a") \\
&\qquad \cdot VCumCO2Capt(runCy, YTIME) \\
&\qquad + iElastCO2Seq(runCy, "mc_b")) \\
&\quad + VWghtTrnstLnrToExpo(runCy, YTIME) \\
&\quad \cdot (iElastCO2Seq(runCy, "mc_c") \\
&\qquad \cdot exp\,(iElastCO2Seq(runCy, "mc_d") \\
&\qquad \cdot VCumCO2Capt(runCy, YTIME)))
\end{aligned}
$$

$$
\begin{aligned}
&VTotGhgEmisAllCountrNap(NAP, YTIME) \\
&= \Bigg(\sum_{runCy} \Bigg(\sum_{(EFS,INDSE)\$(SECTTECH(INDSE,EFS)\$NAPtoALLSBS(NAP,INDSE))} (VConsFuel(runCy, INDSE, E \\
&\quad + \sum_{PGEF} (VTransfInThermPowPls(runCy, PGEF, YTIME) \cdot iCo2EmiFac(runCy, "PG", PGEF, YTIM \\
&\quad + \sum_{EFS} (VTransfInputDHPlants(runCy, EFS, YTIME) \cdot iCo2EmiFac(runCy, "PG", EFS, YTIME)) \\
&\quad + \sum_{EFS} (VEnCons(runCy, EFS, YTIME) \cdot iCo2EmiFac(runCy, "PCH", EFS, YTIME)) \\
&\quad - \sum_{PGEF} \Bigg(\sum_{CCS\$PGALLtoEF(CCS,PGEF)} \Bigg(VElecProd(runCy, CCS, YTIME) \cdot \frac{sTWhToMtoe}{iPlantEffByType(runCy, CC}
\end{aligned}
$$

$$
VTotCo2AllCoun(YTIME) = \sum_{NAP} VTotGhgEmisAllCountrNap(NAP, YTIME)
$$

$$
\begin{aligned}
&VHouseExpEne(runCy, YTIME) \\
&= SUM \Bigg(DSBS\$HOU(DSBS), SUM \Bigg(EF\$SECTTECH(dSBS, EF), VConsRemSubEquip(runCy, DSB \\
&\quad \cdot \Bigg(VFuelPriceSub(runCy, DSBS, EF, YTIME) - \frac{iEffValueInEuro(runCy, DSBS, YTIME)}{1000} - iCo2Er \\
&\quad + VElecPriIndResNoCliPol(runCy, "R", YTIME) \\
&\quad \cdot \frac{VElecNonSub(runCy, "HOU", YTIME)}{sTWhToMtoe}
\end{aligned}
$$

$$
\begin{aligned}
VFuelPriceSub(runCy, SBS, EF, YTIME) = \Bigg(\Bigg(1 \\
+ \sum_{WEF\$EFtoWEF(SBS,EF,WEF)} \Bigg(\sum_{NAP\$NAPtoALLSBS(NAP,SBS)} iScenarioPri(WEF, NAP, YTIME) \Bigg)\Bigg) \\
\cdot ((1 - iPriceReform(runCy, SBS, EF, YTIME)) \\
\cdot VFuelPriceSub(runCy, SBS, EF, YTIME - 1) \\
+ iPriceReform(runCy, SBS, EF, YTIME) \cdot iPriceTragets(runCy, SBS, EF, YTIME)) \\
+ iCo2EmiFac(runCy, SBS, EF, YTIME) \\
\cdot \frac{\sum_{NAP\$NAPtoALLSBS(NAP,SBS)} VCarVal(runCy, NAP, YTIME)}{1000} \\
+ \Bigg(\frac{(iEffValueInEuro(runCy, SBS, YTIME) - iEffValueInEuro(runCy, SBS, YTIME - 1))}{1000} \Bigg) \$DS \\
+ (iFuelPrice(runCy, SBS, EF, \text{"2018"})\$DSBS(SBS)\$ALTEF(EF)) \\
+ \Bigg(\frac{(VElecPriInduResConsu(runCy, \text{"i"}, YTIME)\$INDTRANS(SBS) + VElecPriInduResConsu(run}{sTWhToMtoe} \\
+ \Bigg(\frac{iEffValueInEuro(runCy, SBS, YTIME)}{1000} \Bigg) \$DSBS(SBS) \Bigg) \$\,(ELCEF(EF) or HEATPUMP(EF)) \\
+ (iHydrogenPri(runCy, SBS, YTIME \\
- 1)\$DSBS(SBS))\,\$\,(H2EF(EF) or sameas(\text{"STE1AH2F"}, EF))
\end{aligned}
$$

$$
\begin{aligned}
VFuelPriMultWgt(runCy, DSBS, EF, YTIME) \\
= iWgtSecAvgPriFueCons(runCy, DSBS, EF) \\
\cdot VFuelPriceSub(runCy, DSBS, EF, YTIME)
\end{aligned}
$$

$$
\begin{aligned}
VFuelPriceAvg(runCy, DSBS, YTIME) \\
= \sum_{EF\$SECTTECH(DSBS,EF)} VFuelPriMultWgt(runCy, DSBS, EF, YTIME)
\end{aligned}
$$

$$
\begin{aligned}
VElecPriInduResConsu(runCy, ESET, YTIME) \\
= (1 + iFacElecPriConsu(runCy, \text{"VAT"}, YTIME)) \\
\cdot (((VFuelPriceSub(runCy, \text{"OI"}, \text{"ELC"}, YTIME - 1) \cdot sTWhToMtoe)\,\$TFIRST(YTIME - 1) \\
+ (iFacElecPriConsu(runCy, \text{"IND_RES"}, YTIME - 1) + VRenShareElecProdSub(runCy, YTIME - \\
+ ((VFuelPriceSub(runCy, \text{"HOU"}, \text{"ELC"}, YTIME - 1) \cdot sTWhToMtoe)\,\$TFIRST(YTIME - 1) \\
+ (iFacElecPriConsu(runCy, \text{"TERT_RES"}, YTIME - 1) + VRenShareElecProdSub(runCy, YTIME -
\end{aligned}
$$

$$
vDummyObj = 1
$$

Preloop (preloop.gms)

Description

```
iNPDL(DSBS) = 6;
loop DSBS do
   loop KPDL$(ord(KPDL) le iNPDL(DSBS)) do
         iFPDL(DSBS,KPDL) = 6 * (iNPDL(DSBS)+1-ord(KPDL)) * ord(KPDL)
                            /
                            (iNPDL(DSBS) * (iNPDL(DSBS)+1) * (iNPDL(DSBS
                               )+2))
   endloop;
endloop;
model openprom /
QDemSub
qDummyObj
/;
option iPop:2:0:6;
display iPop;
display iDisc;
display TF;
display TFIRST;
display runCy;
display iWgtSecAvgPriFueCons;
display iTotFinEneDemSubBaseYr;
VNumVeh.L(allCy,YTIME)=0.1;
VTrnspActiv.l(allCy,TRANSE,YTIME) = 0.1;
VFuelPriceSub.l(allCy,SBS,EF,YTIME) = 0.1;
VElecIndPrices.l(allCy,YTIME)= 0.1;
VTechCostVar.l(allCy,SBS,EF,TEA,YTIME) = 0.1;
VTechCostIntrm.l(allCy,DSBS,rcon,EF,TEA,YTIME) = 0.1;
VLifeTimeTech.l(allCy,DSBS,EF,TEA,YTIME)= 0.1;
VTechSort.l(allCy,DSBS,rCon,YTIME) = 0.1;
VConsFuel.l(allCy,DSBS,EF,YTIME)=0.1;
VRefCapacity.l(allCy,YTIME)=0.1;
VFeCons.l(allCy,EF,YTIME) = 0.1;
VTransfOutputRefineries.l(allCy,EFS,YTIME)=0.1;
VGrsInlConsNotEneBranch.l(allCy,EFS,YTIME)=0.1;
VElecConsAll.l(allCy,DSBS,YTIME)=0.1;
VCapChpPlants.l(allCy,YTIME)=0.1;
VElecPeakLoad.l(allCy,YTIME)=0.1;
VElecPeakLoad.up(allCy,YTIME)=1e6;
VElecProdPowGenPlants.l(allCy,PGALL,YTIME) = 1;
```

```
VSensCcs.l(allCy,YTIME)=1;
VCarVal.fx(allCy,NAP,YTIME)=1;
VFuelPriceSub.l(allCy,"PG",PGEF,YTIME)=1;
VProdCostTechnology.l(allCy,PGALL2,YTIME)=0.1;
VProdCostTechnology.up(allCy,PGALL2,YTIME)=1e6;
VVarCostTech.l(allCy,PGALL,YTIME)=0.1;
VProdCostTechPreReplacAvail.l(allCy,PGALL,PGALL2,YTIME)=0.1;
VTechSortVarCost.l(allCy,TRANSE,Rcon,YTIME)=0.1;
VHourProdCostTechNoCCS.up(allCy,PGALL,HOUR,YTIME)=8000;
VRenPotSupplyCurve.l(allCy,PGRENEF,YTIME)=0.1;
VScrRate.l(allCy,YTIME)=0.1;
VTranspCostPermeanConsSize.l(allCy,TRANSE,RCon,TTECH,TEA,YTIME)=0.1;
VTranspCostPermeanConsSize.lo(allCy,TRANSE,RCon,TTECH,TEA,YTIME)
    =0.0001;
VElecNonSub.l(allCy,DSBS,YTIME)=0.1;
VPowerPlantNewEq.L(allCy,PGALL,TT)=0.1;
VHourProdCostTechAfterCCS.L(allCy,PGALL,HOUR,TT)=0.1;
VPowerPlaShrNewEq.L(allCy,PGALL,TT)=0.1;
VHourProdCostTechNoCCS.L(runCy,PGALL,HOUR,TT) = VPowerPlantNewEq.L(
    runCy,PGALL,TT)*VHourProdCostTechAfterCCS.L(runCy,PGALL,HOUR,TT)+
          sum(CCS$CCS_NOCCS(CCS,PGALL), VPowerPlaShrNewEq.L(runCy,CCS,TT
              )*VHourProdCostTechAfterCCS.L(runCy,CCS,HOUR,TT));
VHourProdCostTechNoCCS.SCALE(runCy,PGALL,HOUR,TT)= max(abs(
    VHourProdCostTechNoCCS.L(runCy,PGALL,HOUR,TT)),1E-20);
VNewInvDecis.l(allCy,YTIME)=0.1;
VVarCostTechnology.l(allCy,PGALL,YTIME)=0.1;
VElecPeakLoads.l(allCy,YTIME)=0.1;
VNewCapYearly.l(allCy,PGALL,YTIME)=0.1;
VAvgCapFacRes.l(allCy,PGALL,YTIME)=0.1;
VOverallCap.scale(allCy,PGALL,YTIME)=1;
VPowPlantSorting.l(runCy,PGALL,YTIME)=0.01;
VReqElecProd.l(runCy,YTIME)=0.1;
VPowPlantSorting.up(runCy,PGALL,YTIME)=0.001;
VPowPlantSorting.scale(runCy,PGALL,YTIME)=1;
VElecDem.l(allCy,YTIME)=0.1;
VRenTechMatMult.l(allCy,PGALL,YTIME)=0.1;
VGoodsTranspActiv.l(allCy,TRANSE,YTIME)=0.1;
VRenShareElecProdSub.FX(runCy,YTIME)$(NOT AN(YTIME))=0;
VRenValue.l(YTIME)=1;
VCO2CO2SeqCsts.l(allCy,YTIME)=1;
VScalWeibullSum.l(allCy,PGALL,YTIME)=2;
VHourProdCostTech.l(runCy,PGALL,HOUR,TT) = 0.0001;
```

```
VFuelPriceSub.FX(runCy,SBS,EF,YTIME)$(SECTTECH(SBS,EF) $(not HEATPUMP(EF))  $(not An(YTIME))) = iFuelPrice(runCy,SBS,EF,YTIME);
VFuelPriceSub.FX(runCy,SBS,ALTEF,YTIME)$(SECTTECH(SBS,ALTEF) $(not An(YTIME))) = sum(EF$ALTMAP(SBS,ALTEF,EF),iFuelPrice(runCy,SBS,EF,YTIME));
VFuelPriceSub.FX(runCy,"PG","NUC",YTIME) = 0.025; !! fixed price for nuclear fuel to 25Euro/toe
VFuelPriceSub.FX(runCy,"H2P","NUC",YTIME) = 0.025; !! fixed price for nuclear fuel to 25Euro/toe
VFuelPriceSub.fx(runCy,INDDOM,"HEATPUMP",YTIME)$(SECTTECH(INDDOM,"HEATPUMP") $(not An(YTIME))) = iFuelPrice(runCy,INDDOM,"ELC",YTIME);
VFuelPriceSub.fx(runCy,"H2P",EF,YTIME)$(SECTTECH("H2P",EF) $(not An(YTIME))) = VFuelPriceSub.l(runCy,"PG",EF,YTIME);
VFuelPriceSub.fx(runCy,"H2P","ELC",YTIME)$(not An(YTIME)) = VFuelPriceSub.l(runCy,"OI","ELC",YTIME);
VElecPriInduResConsu.FX(runCy,"i",YTIME)$(not an(ytime)) = VFuelPriceSub.l(runCy,"OI","ELC",YTIME)*0.086;
VElecPriInduResConsu.FX(runCy,"r",YTIME)$(not an(ytime)) = VFuelPriceSub.l(runCy,"HOU","ELC",YTIME)*0.086;
VElecPriIndResNoCliPol.FX(runCy,"i",YTIME)$(not an(ytime)) = VFuelPriceSub.l(runCy,"OI","ELC",YTIME)*0.086;
VElecPriIndResNoCliPol.FX(runCy,"r",YTIME)$(not an(ytime)) = VFuelPriceSub.l(runCy,"HOU","ELC",YTIME)*0.086;
VFuelPriSubNoCarb.FX(runCy,SBS,EF,YTIME)$(SECTTECH(SBS,EF) $(not HEATPUMP(EF))  $(not An(YTIME))) = iFuelPrice(runCy,SBS,EF,YTIME);
VFuelPriSubNoCarb.FX(runCy,SBS,ALTEF,YTIME)$(SECTTECH(SBS,ALTEF) $(not An(YTIME))) = sum(EF$ALTMAP(SBS,ALTEF,EF),iFuelPrice(runCy,SBS,EF,YTIME));
VFuelPriSubNoCarb.FX(runCy,"PG","NUC",YTIME) = 0.025; !! fixed price for nuclear fuel to 25Euro/toe
VFuelPriSubNoCarb.fx(runCy,INDDOM,"HEATPUMP",YTIME)$(SECTTECH(INDDOM,"HEATPUMP") $(not An(YTIME))) = iFuelPrice(runCy,INDDOM,"ELC",YTIME);
VFuelPriceAvg.FX(runCy,DSBS,YTIME) = sum(EF$SECTTECH(DSBS,EF),iWgtSecAvgPriFueCons(runCy,DSBS,EF) * iFuelPrice(runCy,DSBS,EF,YTIME));
VNumVeh.UP(runCy,YTIME) = 10000; !! upper bound of VNumVeh is 10000 million vehicles
VNumVeh.FX(runCy,YTIME)$(not An(YTIME)) = iActv(YTIME,runCy,"PC");
VLamda.UP(runCy,YTIME) = 1;
VLamda.FX(runCy,YTIME)$((not An(YTIME)) $(ord(YTIME) gt 1) ) = (VNumVeh.l(runCy,YTIME-1) / (iPop(YTIME-1,runCy)*1000) /
        iPassCarsMarkSat(runCy))$(iPop(YTIME-1,runCy))+VLamda.l(runCy,YTIME-1)$(not iPop(YTIME-1,runCy));
```

```
VMExtF.l(runCy,YTIME)$((not An(YTIME)) $(ord(YTIME) gt 1)  ) = (iTransChar(runCy,"RES_MEXTF",YTIME) * iSigma(runCy,"S1") * EXP(iSigma(runCy,"S2") *
            EXP(iSigma(runCy,"S3") * VLamda.l(runCy,YTIME)))
                * VNumVeh.l(runCy,YTIME-1) /(iPop(YTIME-1,runCy) * 1000) )$(iPop(YTIME-1,runCy));
VMExtF.FX(runCy,YTIME)$((not An(YTIME)) $(ord(YTIME) gt 1)  ) = (iTransChar(runCy,"RES_MEXTF",YTIME) * iSigma(runCy,"S1") * EXP(iSigma(runCy,"S2") * EXP(iSigma(runCy,"S3") *
                          VLamda.l(runCy,YTIME)))*
                          VNumVeh.l(runCy,YTIME-1) /(iPop(YTIME-1,runCy) * 1000) )$(iPop(YTIME-1,runCy))+VMExtF.l(runCy,YTIME-1)$(not iPop(YTIME-1,runCy));
VMExtV.FX(runCy,YTIME)$(not An(YTIME)) = iDataPassCars(runCy,"PC","MEXTV");
VScrRate.UP(runCy,YTIME) = 1;
VScrRate.FX(runCy,YTIME)$(not An(YTIME)) = iDataPassCars(runCy,"PC","scr");
VGapTranspFillNewTech.FX(runCy,TRANSE,YTIME)$(not AN(YTIME))=0;
VTrnspActiv.FX(runCy,"PC",YTIME)$(not An(YTIME)) = iTransChar(runCy,"KM_VEH",YTIME);
VTrnspActiv.FX(runCy,TRANP,YTIME)$((not An(YTIME)) $(not sameas(TRANP,"PC"))) = iActv(YTIME,runCy,TRANP);
VTrnspActiv.FX(runCy,TRANSE,YTIME)$(not TRANP(TRANSE)) = 0;
VGoodsTranspActiv.FX(runCy,TRANG,YTIME)$(not An(YTIME)) = iActv(YTIME,runCy,TRANG);
VGoodsTranspActiv.FX(runCy,TRANSE,YTIME)$(not TRANG(TRANSE)) = 0;
VLifeTimeTech.FX(runCy,DSBS,EF,TEA,YTIME)$(SECTTECH(DSBS,EF)  $(not TRANSE(DSBS)) $(not sameas(DSBS,"PC"))) = iTechLft(runCy,DSBS,EF,YTIME);
VLifeTimeTech.FX(runCy,TRANSE,TTECH,TEA,YTIME)$(SECTTECH(TRANSE,TTECH) $(not sameas(TRANSE,"PC"))) = iTechLft(runCy,TRANSE,TTECH,YTIME);
VLifeTimeTech.FX(runCy,DSBS,EF,TEA,YTIME)$(not SECTTECH(DSBS,EF))=0;
VLifeTimeTech.FX(runCy,"PC",TTECH,TEA,YTIME)$( (not AN(YTIME)) $SECTTECH("PC",TTECH))=10;
VSpecificFuelCons.FX(runCy,TRANSE,TTECH,TEA,EF,"2017")$(SECTTECH(TRANSE,EF) ) = iSpeFuelConsCostBy(runCy,TRANSE,TTECH,TEA,EF);
VTechSortVarCostNewEquip.FX(runCy,TRANSE,TTECH,TEA,YTIME)$( SECTTECH(TRANSE,TTECH) $(not AN(YTIME))) = 0;
VConsEachTechTransp.FX(runCy,TRANSE,TTECH,EF,TEA,YTIME)$(SECTTECH(TRANSE,TTECH)  $(not PLUGIN(TTECH)) $TTECHtoEF(TTECH,EF)  $(not An(YTIME))) = iFuelConsPerFueSub(runCy,TRANSE,EF,YTIME);
```

```
VConsEachTechTransp.FX(runCy,TRANSE,TTECH,EF,TEA,YTIME)$(SECTTECH(TRANSE,TTECH)  $PLUGIN(TTECH) $(not An(YTIME))) = 0;
VConsEachTechTransp.FX(runCy,TRANSE,TTECH,EF,TEA,YTIME)$(SECTTECH(TRANSE,TTECH)  $CHYBV(TTECH) $(not An(YTIME))) =0;
VDemTr.FX(runCy,TRANSE,EF,YTIME) $(SECTTECH(TRANSE,EF) $(not An(YTIME))) = iFuelConsPerFueSub(runCy,TRANSE,EF,YTIME);
VElecNonSub.FX(runCy,INDDOM,YTIME)$(not An(YTIME)) = iFuelConsPerFueSub(runCy,INDDOM,"ELC",YTIME) * iShrNonSubElecInTotElecDem(runCy,INDDOM);
VElecConsInd.FX(runCy,YTIME)$(not An(YTIME))= SUM(INDSE,VElecNonSub.l(runCy,INDSE,YTIME));
VFuePriSubChp.FX(runCy,DSBS,EF,TEA,YTIME)$((not An(YTIME)) $(not TRANSE(DSBS))  $SECTTECH(DSBS,EF)) =
(((VFuelPriceSub.l(runCy,DSBS,EF,YTIME)+iVarCostTech(runCy,DSBS,EF,YTIME)/1000)/iUsfEneConvSubTech(runCy,DSBS,EF,YTIME)-
(0$(not CHP(EF)) + (VFuelPriceSub.l(runCy,"OI","ELC",YTIME)*iFracElecPriChp(runCy,YTIME)*iElecIndex(runCy,"2010"))$CHP(EF))) + (0.003) +
SQRT( SQR(((VFuelPriceSub.l(runCy,DSBS,EF,YTIME)+iVarCostTech(runCy,DSBS,EF,YTIME)/1000)/iUsfEneConvSubTech(runCy,DSBS,EF,YTIME)- (0$(not CHP(EF)) +
(VFuelPriceSub.l(runCy,"OI","ELC",YTIME)*iFracElecPriChp(runCy,YTIME)*iElecIndex(runCy,"2010"))$CHP(EF)))-(0.003)) + SQR(1e-7) ) )/2;
VDemSub.FX(runCy,INDDOM,YTIME)$(not An(YTIME)) = max(iTotFinEneDemSubBaseYr(runCy,INDDOM,YTIME) - VElecNonSub.L(runCy,INDDOM,YTIME),1e-5);
VDemSub.FX(runCy,NENSE,YTIME)$(not An(YTIME)) = max(iTotFinEneDemSubBaseYr(runCy,NENSE,YTIME),1e-5);
VDemSub.FX(runCy,"HOU",YTIME)$(not An(YTIME)) = max(iTotFinEneDemSubBaseYr(runCy,"HOU",YTIME) - VElecNonSub.L(runCy,"HOU",YTIME)-iExogDemOfBiomass(runCy,"HOU",YTIME),1e-5);
VDemInd.FX(runCy,YTIME)$(not An(YTIME))= SUM(INDSE,VDemSub.L(runCy,INDSE,YTIME));
VElecIndPrices.FX(runCy,YTIME)$TFIRST(YTIME) = iElecIndex(runCy,YTIME);
VElecConsHeatPla.FX(runCy,INDDOM,YTIME)$(not An(YTIME)) = iFuelConsPerFueSub(runCy,INDDOM,"ELC",YTIME)*(1-iShrNonSubElecInTotElecDem(runCy,INDDOM))*iShrHeatPumpElecCons(runCy,INDDOM);
VConsFuel.FX(runCy,DSBS,EF,YTIME)$(SECTTECH(DSBS,EF) $(not HEATPUMP(EF)) $(not TRANSE(DSBS)) $(not An(YTIME))) = iFuelConsPerFueSub(runCy,DSBS,EF,YTIME);
VConsFuelSub.FX(runCy,DSBS,EF,YTIME)$(SECTTECH(DSBS,EF) $(not TRANSE(DSBS)) $(not An(YTIME))) =
```

```
(iFuelConsPerFueSub(runCy,DSBS,EF,YTIME))$((not ELCEF(EF)) $(not HEATPUMP(EF)))
+(VElecConsHeatPla.l(runCy,DSBS,YTIME)*iUsfEneConvSubTech(runCy,DSBS,"HEATPUMP",YTIME))$HEATPUMP(EF)+
(iFuelConsPerFueSub(runCy,DSBS,EF,YTIME)-VElecConsHeatPla.l(runCy,DSBS,YTIME))$ELCEF(EF)
+1e-7$(H2EF(EF) or sameas("STE1AH2F",EF))
;
VTechShareNewEquip.FX(runCy,DSBS,EF,TEA,YTIME)$(SECTTECH(DSBS,EF)$(not An(YTIME))) = 0;
VConsRemSubEquip.FX(runCy,DSBS,EF,YTIME)$(SECTTECH(DSBS,EF) $(not An(ytime))) =
(VConsFuelSub.l(runCy ,DSBS,EF,YTIME) - (VElecNonSub.l(runCy,DSBS,YTIME)$(ELCEF(EF) $INDDOM(DSBS)) + 0$(not (ELCEF(EF) $INDDOM(DSBS)) )))
;
VFeCons.FX(runCy,EFS,YTIME)$(not An(YTIME)) = iFinEneCons(runCy,EFS,YTIME);
iFinEneConsPrevYear(runCy,EFS,YTIME)$(not An(YTIME)) = iFinEneCons(runCy,EFS,YTIME);
VLosses.FX(runCy,EFS,YTIME)$(not An(YTIME)) = iDistrLosses(runCy,EFS,YTIME);
VTransfOutputPatFuel.FX(runCy,EFS,YTIME)$(not An(YTIME)) = iTranfOutGasworks(runCy,EFS,YTIME);
VTransfOutThermPowSta.FX(runCy,EFS,YTIME)$(not TOCTEF(EFS)) = 0;
VRefCapacity.FX(runCy,YTIME)$(not An(YTIME)) = iRefCapacity(runCy,YTIME);
VTransfOutputRefineries.FX(runCy,EFS,YTIME)$(EFtoEFA(EFS,"LQD") $(not An(YTIME))) = iTransfOutputRef(runCy,EFS,YTIME);
VTransfInputRefineries.FX(runCy,"CRO",YTIME)$(not An(YTIME)) = iTransfInputRef(runCy,"CRO",YTIME);
VGrsInlConsNotEneBranch.FX(runCy,EFS,YTIME)$(not An(YTIME)) = iGrossInConsNoEneBra(runCy,EFS,YTIME);
VGrssInCons.FX(runCy,EFS,YTIME)$(not An(YTIME)) = iGrosInlCons(runCy,EFS,YTIME);
VTransfers.FX(runCy,EFS,YTIME)$(not An(YTIME)) = iFeedTransfr(runCy,EFS,YTIME);
VPrimProd.FX(runCy,PPRODEF,YTIME)$(not An(YTIME)) = iFuelPriPro(runCy,PPRODEF,YTIME);
VEnCons.FX(runCy,EFS,YTIME)$(not An(YTIME)) = iTotEneBranchCons(runCy,EFS,YTIME);
VExportsFake.FX(runCy,EFS,YTIME)$(not An(YTIME)) = iFuelExprts(runCy,EFS,YTIME);
```

```
VFkImpAllFuelsNotNatGas.FX(runCy,"NGS",YTIME)$(not An(YTIME)) = iFuelImports(runCy,"NGS",YTIME);
VExportsFake.FX(runCy,"NGS",YTIME)$(not An(YTIME)) = iFuelExprts(runCy,"NGS",YTIME);
VElecDem.FX(runCy,YTIME)$(not An(YTIME)) =  1/0.086 * ( iFinEneCons(runCy,"ELC",YTIME) + sum(NENSE, iFuelConsPerFueSub(runCy,NENSE,"ELC",YTIME)) + iDistrLosses(runCy,"ELC",YTIME)
                                                    + iTotEneBranchCons(runCy,"ELC",YTIME) - (iFuelImports(runCy,"ELC",YTIME)-iFuelExprts(runCy,"ELC",YTIME)));
VCorrBaseLoad.FX(runCy,YTIME)$(not An(YTIME)) = iPeakBsLoadBy(runCy,"BASELOAD");
$ontext
VCapChpPlants.FX(runCy,YTIME)$(datay(YTIME)) =
         (sum(INDDOM,VConsFuel.l(runCy,INDDOM,"ELC",YTIME)) + sum(TRANSE, VDemTr.l(runCy,TRANSE,"ELC",YTIME)))/
         (sum(INDDOM,VConsFuel.l(runCy,INDDOM,"ELC",YTIME)/iLoadFacElecDem(runCy,INDDOM,"2015")) + sum(TRANSE, VDemTr.l(runCy,TRANSE,"ELC",YTIME)/
         iLoadFacElecDem(runCy,TRANSE,"2015")));
$offtext
VElecPeakLoad.FX(runCy,YTIME)$(datay(YTIME)) = VElecDem.l(runCy,YTIME)/(VCapChpPlants.l(runCy,YTIME)*8.76);
VTotElecGenCap.FX(runCy,YTIME)$(not An(YTIME)) = iTotAvailCapBsYr(runCy);
VElecGenNoChp.FX(runCy,YTIME)$(not An(YTIME)) = iTotAvailCapBsYr(runCy);
VElecCapChpPla.FX(runCy,CHP,YTIME)$(not An(YTIME)) = iHisChpGrCapData(runCy,CHP,YTIME);
VPowPlaShaNewEquip.FX(runCy,PGALL,YTIME)$((NOT AN(YTIME)) )=0;
VHourProdCostTech.FX(runCy,PGALL,HOUR,YTIME)$((NOT AN(YTIME)))=0;
VPowerPlaShrNewEq.FX(runCy,PGALL,YTIME)$((NOT AN(YTIME)) )=0;
VPowerPlantNewEq.FX(runCy,PGALL,YTIME)$((NOT AN(YTIME)) )=0;
VPowerPlaShrNewEq.FX(runCy,PGALL,YTIME)$(AN(YTIME) $(NOT CCS(PGALL)))=0;
VPowerPlantNewEq.FX(runCy,PGALL,YTIME)$(AN(YTIME) $(NOT NOCCS(PGALL)) )=0;
VNewInvDecis.FX(runCy,YTIME)$(NOT AN(YTIME))=1;
VElecGenPlanCap.FX(runCy,PGALL,YTIME)$DATAY(YTIME) =  iInstCapPast(runCy,PGALL,YTIME);
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```
VElecGenPlantsCapac.FX(runCy,PGALL,YTIME)$DATAY(YTIME) =  iInstCapPast(runCy,PGALL,YTIME);
VOverallCap.FX(runCy,PGALL,YTIME)$TFIRST(YTIME) =  iInstCapPast(runCy,PGALL,YTIME)$TFIRST(YTIME);
VNewCapYearly.FX(runCy,PGALL,"2011")$PGREN(PGALL) = iInstCapPast(runCy,PGALL,"2011")- iInstCapPast(runCy,PGALL,"2010") +1E-10;
VNewCapYearly.FX(runCy,PGALL,"2012")$PGREN(PGALL) = iInstCapPast(runCy,PGALL,"2012")- iInstCapPast(runCy,PGALL,"2011") +1E-10;
VNewCapYearly.FX(runCy,PGALL,"2013")$PGREN(PGALL) = iInstCapPast(runCy,PGALL,"2013")- iInstCapPast(runCy,PGALL,"2012") +1E-10;
VNewCapYearly.FX(runCy,PGALL,"2014")$PGREN(PGALL) = iInstCapPast(runCy,PGALL,"2014")- iInstCapPast(runCy,PGALL,"2013") +1E-10;
VNewCapYearly.FX(runCy,PGALL,"2015")$PGREN(PGALL) = iInstCapPast(runCy,PGALL,"2015")- iInstCapPast(runCy,PGALL,"2014") +1E-10;
VNewCapYearly.FX(runCy,PGALL,"2016")$PGREN(PGALL) = iInstCapPast(runCy,PGALL,"2016")- iInstCapPast(runCy,PGALL,"2015") +1E-10;
VNewCapYearly.FX(runCy,PGALL,"2017")$PGREN(PGALL) = iInstCapPast(runCy,PGALL,"2017")- iInstCapPast(runCy,PGALL,"2016") +1E-10;
VNewCapYearly.FX(runCy,"PGLHYD",YTIME)$TFIRST(YTIME) = +1E-10;
VAvgCapFacRes.FX(runCy,PGALL,YTIME)$DATAY(YTIME) =iAvailRate(PGALL,YTIME);
VElecProdPowGenPlants.FX(runCy,pgall,YTIME)=iDataElecProd(runCy,pgall,YTIME)/1000;
VEndogScrapIndex.FX(runCy,PGALL,YTIME)$(not an(YTIME) ) = 1;
VEndogScrapIndex.FX(runCy,PGSCRN,YTIME) = 1;            !! premature replacement it is not allowed for all new plants
VCO2ElcHrgProd.FX(runCy,YTIME)$(not An(YTIME)) = 0;
VRenShareElecProdSub.FX(runCy,YTIME)$(NOT AN(YTIME))=0;
VRenPotSupplyCurve.FX(runCy,PGRENEF, YTIME) $(NOT AN(YTIME)) = iMinRenPotential(runCy,PGRENEF,YTIME);
VAvgPowerGenLongTrm.L(runCy,ESET,"2010") = 0;
VLonPowGenCostNoClimPol.L(runCy,ESET,"2010") = 0;
VAvgPowerGenCostShoTrm.L(runCy,ESET,"2010") = 0;
VLongPowGenCost.L(runCy,PGALL,ESET,"2010") = 0;
VLonAvgPowGenCostNoClimPol.L(runCy,PGALL,ESET,"2010") = 0;
VAvgPowerGenLongTrm.L(runCy,ESET,"2017") = 0;
VLonPowGenCostNoClimPol.L(runCy,ESET,"2017") = 0;
VAvgPowerGenCostShoTrm.L(runCy,ESET,"2017") = 0;
VLongPowGenCost.L(runCy,PGALL,ESET,"2017") = 0;
VLonAvgPowGenCostNoClimPol.L(runCy,PGALL,ESET,"2017") = 0;
VCarVal.fx(runCy,NAP,YTIME)$(not An(YTIME))=0;
VCarVal.FX(runCy,"TRADE",YTIME)$an(YTIME) = sExogCarbValue*iCarbValYrExog(YTIME);
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VCarVal.FX(runCy,"NOTRADE",YTIME)$an(YTIME) =sExogCarbValue*iCarbValYrExog(YTIME);
VCumCO2Capt.FX(runCy,YTIME)$(not an(YTIME)) = 0 ;
VCO2CO2SeqCsts.FX(runCy,YTIME)$(not an(YTIME)) =
       (iElastCO2Seq(runCy,"mc_a") *iElastCO2Seq(runCy,"mc_b"));
VDemTr.FX(runCy,TRANSE,EF,YTIME)$(not SECTTECH(TRANSE,EF)) = 0;
VTransfOutputDHPlants.FX(runCy,EFS,YTIME)$(not STEAM(EFS)) = 0;
VTransfOutputRefineries.FX(runCy,EFS,YTIME)$(not EFtoEFA(EFS,"LQD")) = 0;
VTransfInputRefineries.FX(runCy,EFS,YTIME)$(not sameas("CRO",EFS)) = 0;
VTransfOutputNuclear.FX(runCy,EFS,YTIME)$(not sameas("ELC",EFS)) = 0;
VTransfInNuclear.FX(runCy,EFS,YTIME)$(not sameas("NUC",EFS)) = 0;
VTransfInThermPowPls.FX(runCy,EFS,YTIME)$(not PGEF(EFS)) = 0;
VExportsFake.FX(runCy,EFS,YTIME)$(not IMPEF(EFS)) = 0;
VFkImpAllFuelsNotNatGas.FX(runCy,EFS,YTIME)$(not IMPEF(EFS)) = 0;
VScalFacPlaDisp.LO(runCy, HOUR, YTIME)=-1;
VLoadCurveConstr.LO(runCy,YTIME)=0;
VLoadCurveConstr.L(runCy,YTIME)=0.01;
VRenValue.FX(YTIME) = 0 ;
VTotReqElecProd.fx(runCy,YTIME)$TFIRST(YTIME)=sum(pgall, VElecProdPowGenPlants.L(runCy,pgall,YTIME)$TFIRST(YTIME));
display VCarVal.l;
execute_unload "blabla.gdx";
loop an do
   s = s + 1;
   TIME(YTIME) = NO;
   TIME(AN)$(ord(an)=s) = YES;
   display TIME;
```

References